Supplemental Materials for Long Read RNA Sequencing in Primary Lung Cell Types Reveals Principles of Nonsense-Mediated Decay

Yul Leshem[1], Yoshiyuki Kasai[2], Masih Eskandar[3], Sanika M. Thakur[4,5], Jay Cabrera[1], Ayan Paul[4,5], John Ziniti[1], Adel Boueiz[1,6], Aabida Saferali[1], Margherita DeMarzio[1], Jennifer Dy[3], Alain Laederach[7], Aleksandra Tata[8], Purushothama Rao Tata[8], Scott H. Randell[2], Peter J. Castaldi[1,9]

[1]Channing Division of Network Medicine, Brigham & Women's Hospital, Harvard Medical School, Boston, MA, USA
[2]Marsico Lung Institute, University of North Carolina at Chapel Hill, Chapel Hill, North Carolina, USA
[3]Department of Electrical and Computer Engineering, Northeastern University, Boston, MA, USA
[4]The Institute for Experiential AI, Northeastern University, Boston, MA, USA
[5]Khoury College of Computer Sciences, Northeastern University, Boston, MA, USA
[6]Division of Pulmonary and Critical Care Medicine, Brigham and Women's Hospital, Boston, MA
[7]Department of Biology, University of North Carolina at Chapel Hill, Chapel Hill, North Carolina, USA
[8]Department of Cell Biology, Duke University, Durham, NC, USA
[9]Division of Primary Care and General Internal Medicine, Brigham & Women's Hospital, Harvard Medical School, Boston, MA, USA

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Supplemental Methods

<u>Isolation and Culturing of Primary Lung Cell Types:</u>

Human lung tissue not used for transplant was obtained, with informed consent, under protocol 03-1396, approved by the University of North Carolina at Chapel Hill Biomedical Institutional Review Board. LAE cells were obtained as previously described[1,2]. MV and FB cells were also obtained from the same donors as previously described[3]. MV, FB, and AT2 cells were obtained from the same three donors. LAE cells were obtained from four different donors and were cultured at 37C, 5% CO_2. MV cells were obtained by dispase and elastase digestion of peripheral lung tissue with the visceral pleura removed and were cultured on type I/III collagen-coated culture wells in EGM-2 media (Lonza, CC-3162) supplemented with FBS. Cells were enriched through two to three rounds of CD31 bead purification (Dynabeads; Life Technologies/ Invitrogen), achieving >95% CD31 positivity by flow cytometry, and were used between passages 5 and 10. FB cells were derived by outgrowth from finely minced distal lung tissue plated onto scratched type I/III collagen-coated dishes in Dulbecco's modified Eagle medium with high glucose (DMEMH) with 10% FBS, antibiotics, and antimycotics. After initial outgrowth, cells were passaged with trypsin/EDTA and sub-cultured under the same conditions. Fibroblast identity was confirmed by morphology and cells were negative for CD31 and pan-cytokeratin which was assessed via flow cytometry and immunofluorescence staining. Additionally, passage one LAE cells procured as described[1], were cultured in modified SAGM complete medium (CC3119, Lonza), supplemented with 1 µM A8301, 1 µM DMH-1, 1 µM CHIR99021, 10 µM Y-27632, and penicillin/streptomycin. AT2 cells were obtained by tissue mincing, enzymatic digestion, and HT2-280 antibody magnetic bead purification as described previously[4]. Cells were cultured and expanded as alveolospheres in Matrigel pellets to passage number 2-3. Alveolospheres were passaged and cultured as monolayer on 5% Matrigel followed by SMG1i or DMSO treatment.

All cultured cells were incubated with 0.3uM of SMG1i or an equivalent volume of DMSO (control), each in their respective media, and cultured at 37C, 5% CO_2 for six hours. After LAE, MV and FB cell cultures were washed with PBS, cells were lysed in RLT Plus buffer (RNeasy Plus Mini Kit, Qiagen 74134). After SMG1i or DMSO treatment AT2 cells were washed once with PBS and lysed in RLT Plus buffer. All lysates were stored at -80C until shipping on dry ice.

Short-read RNA Sequencing

cDNA libraries were constructed from polyA-selected, paired end stranded libraries using the NEBNext ULTRAEXPRESS® RNA Library Prep Kit and sequenced on a Illumina NovaSeq 6000 at the Maryland Genomics core sequencing facility. The mean RIN value for processed samples was 9.3. Libraries were sequenced to an average depth of 59 million reads. RNA-seq data were processed through the nf-core RNA sequencing pipeline (https://nf-co.re/rnaseq/) version 3.14.0 using Nextflow 24.04.4. Read quality and trimming was performed with Trim Galore 0.6.7 (cutadapt 4.6) with default parameter values. Alignment quality and other QC metrics were generated and evaluated with FastQC 0.12.1, samtools 1.17, and MultiQC. Stranded reads were aligned to the human GRCh38 reference genome (GRCh38.primary_assembly.genome.fa.gz) using the STAR aligner 2.7.10a in two-pass mode and the GENCODE 49 GTF (gencode.v49.primary_assembly.annotation.gtf) with default parameters. Junctional reads were assigned according to default parameters. For gene and isoform level quantification, isoform counts were estimated with Salmon version 1.10.1 and default parameters. Gene and isoform level counts were generated with the tximport Bioconductor package (bioconductor-tximeta 1.12.0, r-base 4.1.3). The counts used for this analysis were generated with tximport setting the parameter *counts from abundance* = “length scaled TPM”.

Long-read RNA Sequencing

cDNA PacBio Kinnex libraries were constructed from isolated RNA according to PacBio protocol 103-238-700 REV07 with the following modifications: 1.0X SMRTbell cleanup beads were used instead of 0.9X to avoid enrichment of long transcripts in order to achieve properly formed concatenated libraries. Prior to concatenation, each of the eight Kinnex PCR reactions was pooled by equal mass based on their Qubit concentration, instead of including a fixed volume from each tube. The Kinnex FL RNA libraries were sequenced on a PacBio Revio instrument with Revio SMRT cells and SPRQ chemistry to an average depth of 13 million aligned FLNC reads. Long reads were aligned to the human genome (GRCh38.primary_assembly.genome.fa) using minimap2 (minimap2 -ax splice:hq --junc-bed known_junctions.bed --MD --cs=long -uf GRCh38_splice.mmi input.fastq.gz | samtools sort -o out.bam)[5]. Quantification was performed using the PacBio Isocall pipeline (https://github.com/PacificBiosciences/isocall). Long-read isoform assemblies from isocall were classified with SQANTI3 QC v5.5.4[6]. Because TransDecoder2 dominates runtime, we ran SQANTI3 QC in parallel across chromosomes via a custom Nextflow pipeline (https://github.com/peter4244/sqanti3_by_chromosome) that removes malformed-exon transcripts, splits the isocall GTF by chromosome, and invokes sqanti3_qc.py per chromosome using the following command structure: sqanti3_qc.py --isoforms <chrom>.gtf --refGTF gencode.v49.primary_assembly.annotation.gtf --refFasta GRCh38.primary_assembly.genome.fa --CAGE_peak refTSS_v3.3_human_coordinate.hg38.sorted.updated.bed --polyA_motif_list polyA.list.txt --fl_count nmd_isocall.count_matrix.txt --report skip --novel_gene_prefix <chrom> -t <cpus> -o <chrom> -d . SQANTI3 filter and rescue modules were not applied.

Platform Correlation

Long-read isoform counts were aggregated to the gene level by summing over the transcript→gene_id annotation map, and both platforms were reduced to Ensembl gene identifiers with version suffixes removed. The comparison was restricted to the 26 donor-matched samples common to both platforms. The common gene set (n = 29,185) comprised genes with nonzero counts in either platform, excluding only genes with zero counts in both. Each platform was TMM-normalized separately, and per-sample Pearson and Spearman correlations were computed on log2-CPM values (edgeR::cpm, prior.count = 2), summarized both overall and per cell type.

Isoform Landscape Characterization

Characterization of the long-read isoform landscape was performed on DMSO samples only (n = 162,800, see filtering description in Differential Gene and Transcript Expression section below). An isoform was considered expressed in a given cell type if it had a summed count ≥1 across that cell type's DMSO samples. Cell type-restricted isoforms were defined as those detected in only one of the four cell types under this criterion. Pairwise isoform-set overlap across cell types was quantified using the Jaccard index on binary detection vectors. Pairwise quantitative similarity was quantified using Spearman and Pearson correlations of mean log2-CPM values (computed with edgeR::cpm using log = TRUE and prior.count = 1) across isoforms for each cell type under DMSO.

To identify isoforms whose baseline expression distinguishes the four cell types, we performed a one-vs-rest differential expression analysis on DMSO samples only. The long-read isoform DGEList and the short-read gene DGEList were each restricted to DMSO samples from AT2, LAE, FB, and MV, and a DMSO-restricted filterByExpr was re-applied (min.count = 5, min.total.count = 10) using the cell type design matrix, retaining 105,938 isoforms and 21,547 genes for testing. Library sizes were re-normalized via TMM, and voom transformation was applied. The design formula was ~ 0 + cell_type without a treatment term (DMSO being the only condition). Donor was incorporated as a duplicateCorrelation blocking; the estimated within-donor correlation was 0.026. For each cell type, a one-vs-rest contrast was constructed as that cell type minus the mean of the other three (e.g., LAE − (AT2 + FB + MV)/3), fit via contrasts.fit, and tested with empirical Bayes moderation via eBayes. Isoforms or genes with BH-adjusted $p < 0.05$ and a positive contrast coefficient were classified as significantly associated with a given cell type. To borrow strength across the four one-vs-rest contrasts, which are non-independent by construction, we additionally applied multivariate adaptive shrinkage (mashr) to the limma contrast estimates following the same procedure described below for the SMG1i-vs-DMSO analyses. Effect sizes were taken from contrast coefficients and standard errors from coefficient/moderated-t ratios. The null correlation matrix $\hat{v}$ is estimated from a 20,000-feature random subset using estimate_null_correlation_simple. Covariance matrices included both data-driven (PCA-based on strong signals from univariate mash_1by1 at lfsr < 0.05, up to 5 PCs) and canonical structures via cov_canonical. Posterior mean log2 fold changes and local false sign rates were extracted, and features with lfsr < 0.05 and posterior mean log2 fold change > 1 were classified as cell type markers.

Differential Gene and Transcript Expression

To identify genes and isoforms regulated by NMD, we performed differential expression analysis at the gene level using both short-read data, and at the isoform level using long-read data. The analysis was restricted to four primary lung cell types: AT2, LAE, FB, and MV. For all analyses, raw counts were organized into DGEList objects using the edgeR Bioconductor package. Low expressed features were removed using filterByExpr with a minimum count threshold of 5 and a minimum total count of 10 with the full model matrix (~ cell_type + treatment + cell_type:treatment) supplied as the design argument. Library sizes were normalized using the trimmed mean of M-values (TMM) method. After filtering and normalization, the short-read gene-level analysis tested 25,955 genes, and the long-read isoform-level analysis tested 162,800 isoforms. For each analysis, we fit a linear model using limma-voom with the design formula ~ cell_type + treatment + cell_type:treatment, where LAE was used as the reference level for cell type and treatment was SMG1i versus DMSO. To account for the paired donor structure of the experiment (each donor contributing matched SMG1i and DMSO samples), we estimated within-donor correlation using duplicateCorrelation and incorporated this as a blocking factor in the model fit via lmFit. Cell type-specific treatment effects were estimated by extracting contrasts for the SMG1i effect within each cell type using contrasts.fit, followed by empirical Bayes moderation via eBayes.

We then applied multivariate adaptive shrinkage (mashr) [7] to the limma-voom contrast estimates. For each analysis level, effect sizes (log2 fold changes) were taken directly from the contrast coefficient matrix, and standard errors were derived from the ratio of coefficients to moderated t-statistics. Features with non-finite or non-positive standard errors in any cell type were excluded. The null correlation structure across cell types was estimated from a random subset of 20,000 features (seed 42) using estimate_null_correlation_simple. Covariance matrices were specified using both data-driven from cov_pca (npc = min(5, number of cell types − 1)) on strong signals identified by univariate mash_1by1 analysis at lfsr < 0.05 and canonical covariance structures via cov_canonical (cov_ed not applied). The mashr model was fit using the mash function with both sets of covariance matrices. Posterior mean log2 fold changes and local false sign rates (lfsr) were extracted for downstream analysis. Genes or isoforms with lfsr < 0.05 and posterior mean logFC > 0 were classified as NMD susceptible, indicating transcripts whose expression increases upon NMD inhibition and are therefore subject to degradation by NMD under normal conditions.

Cell Type Specificity

To assess the cell type specificity of NMD, every NMD susceptible feature (lfsr < 0.05, posterior mean logFC > 0) was classified as either shared (NMD susceptible in at least one other cell type at the same threshold) or cell type-specific (NMD susceptible in only that cell type). To quantify the similarity of NMD effect sizes across cell types, we computed pairwise Pearson correlations of mashr posterior mean logFC values among NMD susceptible features.

Gene set enrichment analysis (GSEA) was performed on the short-read gene-level mashr posterior mean logFC rankings using fgsea with MSigDB pathway collections (Hallmark, KEGG, Reactome, GO-BP). For each cell type, genes were ranked by signed posterior mean logFC and tested against pathway gene sets with a minimum size of 15 and maximum size of 500. Pathways with FDR < 0.05 and NES > 0 were considered significant.

GSEA was also performed at long-read isoform resolution. Because MSigDB pathway sets are defined at the gene level, each gene was assigned the signed mashr posterior mean log2 fold change of its largest-effect isoform (the isoform with the largest absolute logFC) and genes were ranked by this isoform-anchored statistic. fgsea was then run against the same MSigDB collections (Hallmark, KEGG, Reactome, GO-BP) with identical gene-set size limits (minimum 15, maximum 500) and the same FDR < 0.05 and NES > 0 thresholds, separately for each cell type.

<u>Transcriptional Output Lost and Productive Isoform Response</u>

To quantify the overall transcriptional burden of NMD, we developed a measure termed percent transcriptional output lost. For each cell type and donor, the value was computed as the summed positive expression differences (SMG1i minus DMSO) across all isoforms divided by the total SMG1i expression where $\Delta_{i,d}^{+} = \max(CPM^{SMG1i} - CPM^{DMSO}, 0)$ and D_C is the set of matched donors for cell type, c. The same measure was computed at the gene level using gene-aggregated CPM values. We also ran the measure with varied CPM floors **(SF22)**. Per-isoform fraction lost was computed as (SMG1i CPM − DMSO CPM) / SMG1i CPM for each NMD susceptible isoform with $\Delta_{i,d}^{+}$.

$$\% \text{ Output Lost}_c = 100 \bullet \frac{\sum_{i=1}^{n} \sum_{d \epsilon D_c} \Delta_{i,d}^{+}}{\sum_{i=1}^{n} \sum_{d \epsilon D_c} CPM_{i,d}^{SMG1i}}$$

To measure how the productive (non-NMD) isoforms of a gene respond when its NMD susceptible isoforms accumulate, we restricted to protein-coding genes (≥1 SQANTI3 annotated coding isoform) and, within each cell type, partitioned each gene's isoforms into a productive compartment and an unproductive compartment. Specifically, the unproductive compartment quantification for each gene in each sample was the sum of expression of all of the NMD susceptible isoforms (mashr lfsr < 0.05, posterior mean log2FC > 0). At the gene level, genes were also labeled as NMD genes (containing ≥1 confident unproductive isoform), non-NMD genes (lfsr>0.2), or uncertain (borderline-up at 0.05 ≤ lfsr < 0.20 or lfsr < 0.05 and log2FC < 0).

Because the accumulation of unproductive isoforms under SMG1i artificially deflates the counts for every productive isoform, we applied a custom normalization that removes this artifact. For gene g with productive isoforms prod(g) and unproductive isoforms unprod(g), the adjustment proceeds in three steps. First, the productive compartment under DMSO was summed:

$$P_g^{DMSO} = \sum_{i \,\epsilon\, prod(g)} \overline{CPM}_i^{DMSO}$$

Second, each unproductive isoform was assigned to the proportion it occupied within that gene under DMSO for each sample, s, and scaled to the current SMG1i productive compartment of that gene.

$$share_i = \frac{\overline{CPM}_i^{DMSO}}{P_g^{DMSO}},\ i \in prod(g)$$

$$\overline{CPM}_{i,s} = share_i \sum_{j \in prod(g)} CPM_{j,s}^{SMG1i}$$

Finally, each adjusted library size was rescaled to one million.

$$CPM_{i,s}^{adj} = \overline{CPM}_{i,s} \bullet \frac{10^6}{\sum_k \overline{CPM}_{k,s}}$$

Unproductive isoforms lacking a positive DMSO productive compartment were held at their DMSO mean. Summing CPM_adj over a gene's productive isoforms gives the adjusted productive CPM, prodCPM_SMG1i_adj, in which the contribution of the unproductive surge has been removed. Throughout, prodCPM and unprodCPM denote the per-gene sums of CPM over prod(g) and unprod(g), totalCPM = prodCPM + unprodCPM, and the superscripts DMSO, SMG1i_raw, and SMG1i_adj denote values under DMSO, uncorrected SMG1i, and NMD-anchored-adjusted SMG1i, respectively. We verified that this normalization left control (non-NMD) genes essentially unchanged between DMSO and adjusted SMG1i (Spearman ≥ 0.96, slope ≈ 1) and preserved gene rankings relative to the uncorrected analysis (Spearman ≥ 0.99 between adjusted and raw fold changes).

From these compartments we derived two per-gene, per-cell-type quantities (means across donors), writing prodCPM and totalCPM for the productive and total gene CPM and f = prodCPM/totalCPM for the productive fraction:

$$\text{Adjusted Productive logFC} = log_2\left(\frac{prodCPM^{SMG1i,\,adj} + 1}{prodCPM^{DMSO} + 1}\right)$$

$$\text{Productive-only Output Shift} = \frac{prodCPM^{SMG1i,\,adj} - prodCPM^{DMSO}}{CPM^{DMSO}}$$

Productive direction was tested per cell type with limma-trend (eBayes) on the adjusted productive logFC, with treatment as the predictor and within-donor correlation incorporated via duplicateCorrelation, then jointly shrunk across cell types with mashr (data-driven plus canonical covariances; null correlation from a random feature subset) exactly as for the primary analyses. Genes with lfsr < 0.05 and posterior mean > 0 (< 0) were called productive-up (productive-down).

To determine whether the productive response analysis captured reproducible signal rather than noise, we performed a series of donor- and platform-based tests operating on the donor-paired adjusted productive log2 fold changes. The Rademacher sign-flip permutation test built the "direction is random" null by flipping donor signs 2,000 times and comparing the number of consistent within-donor effects to the permutation null [8], summarized as an empirical false-discovery rate (expected/observed). Cross-donor sign concordance was compared to its exact binomial chance expectation, 2(1/2)^n for n donors (n=3 or 4). The fraction of genes carrying a real effect was estimated as 1 − π0 with Storey's estimator (tuning parameter λ = 0.5) [9].

We used logistic regression to ask what distinguishes the direction of a gene's productive response. Among responsive genes, we modeled the log-odds that a gene's productive fraction moved up rather than down as a function of: NMD status; baseline total expression (log10 CPM); unproductive-isoform expression at DMSO and at SMG1i (log10 CPM); the SMG1i-induced gain in unproductive expression; the number of isoforms per gene (log10); and short-read gene induction. Cell type was included as a nuisance covariate. Separately, to ask whether a gene responded at all in a given direction, we fit one-versus-rest logistic regressions on the same predictors. The predictive contribution of induction versus NMD status was quantified by a five-fold cross-validated area under the ROC curve[10] for the full model, an induction-only model, an all-features-without-induction model, and an NMD-status-only model. Over-representation of productive-up and productive-down genes was tested with topGO[11] using the elim algorithm and Fisher's exact test and with one-sided Fisher tests against curated MSigDB Hallmark and Gene Ontology sets, using all genes passing the productive-analysis filter as the background.

<u>Productive Response Change Level vs Composition</u>

To separate change in total transcript output from change in isoform composition, the raw productive CPM change, ΔP, was decomposed following the method proposed by Kitagawa [12] into a level (output) term and a composition (isoform-mix) term:

$$\Delta f = f^{SMG1i} - f^{DMSO}$$

$$\Delta P = prodCPM^{SMG1i} - prodCPM^{DMSO}$$

$$\Delta T = CPM^{SMG1i} - CPM^{DMSO}$$

$$\Delta P = \overline{f} \times \Delta T + \overline{T} \times \Delta f$$

where $\overline{f}$ and $\overline{T}$ represent the midpoint between raw SMG1i and DMSO; the additive identity was confirmed numerically. Each gene was assigned to the level- or composition-dominated class by the larger of the two terms in absolute value (|level| vs |composition|). This classification was computed on NMD-target genes with a significant productive direction call (lfsr < 0.05) and reported separately for productive-up and productive-down genes. Non-NMD and uncertain genes, which lack an unproductive compartment and therefore have a composition term of zero by construction, were excluded.

The same causal distinction was found using the custom normalization. Significant productive up and down genes (AT2, LAE, MV) were partitioned by mechanism using the short-read gene induction L, the short-read gene-level log2 fold change, and the productive-fraction change Δf: transcriptional repression when L < −0.2 (total transcription fell, productive isoforms decreasing in parallel), composition shift when Δf < −0.10 with L ≥ −0.1 (transcripts redistributed into NMD-targeted isoforms with little change in total transcription), and mixed otherwise. Because L is a short-read gene-level estimate not derived from the long-read library, this classification provides a transcriptional signal independent of the isoform-level data used in the decomposition. The functional themes of the level and composition groups were compared by one-sided Fisher tests against curated Gene Ontology and Hallmark sets and by topGO over-representation against the same background. Example genes (*SHMT2, SRSF2, PCNA*) were rendered at isoform resolution with Isopair.

RNA-binding Protein and SR Protein Enrichment Among NMD Targets

To test whether NMD preferentially targets RNA-binding proteins (RBPs), we defined an NMD-target gene as any gene with at least one NMD susceptible isoform (long-read isoform-level mashr lfsr < 0.05 and posterior mean log2FC > 0) in at least one cell type, with the background comprising all genes possessing an isoform in the long-read DIE/mashr set. NMD-target genes were intersected with the Gerstberger, Hafner & Tuschl (2014) census of 1,542 human RBPs, matched on Ensembl gene identifier or HGNC symbol, and enrichment was assessed by one-sided Fisher's exact test against the background, both pooled and per cell type. As an orthogonal, experimentally defined reference, the test was repeated against the ENCODE eCLIP roster of RNA-binding proteins (Van Nostrand et al. 2020). Within each census we additionally tested enrichment of individual RBP functional classes (including splicing regulation, core spliceosome, exon junction complex, 3′-end processing, and RNA stability) among NMD targets, using the same one-sided Fisher framework against the DIE-tested background.

The canonical serine/arginine-rich splicing factors (*SRSF1*–*SRSF12*) were checked for NMD susceptibility across the four cell types. To determine the splicing event responsible for the NMD susceptibility of each SR protein, we applied the Isopair framework (described below) to the canonical SR-protein genes *SRSF1*–*SRSF12*. For each SR gene carrying an NMD susceptible isoform, we paired its most highly expressed non-NMD isoform in DMSO (the reference) with its most highly expressed NMD susceptible isoform in SMG1i (the comparator) and enumerated the splice events distinguishing them. The NMD-triggering event was classified as a poison cassette exon, an intron-retention event (including alternatively spliced 3′-UTR introns), or loss of an internal exon, by intersecting each event's genomic coordinates with the comparator's premature termination codon. Premature termination codons were called under the canonical 50-nt rule from each isoform's primary open reading frame. Candidate ORFs were enumerated by a Kozak-aware scan (Isopair::enumerateOrfs), and the primary ORF was selected (Isopair::selectPrimaryOrf) by reference-AUG tracing – projecting the start codon of the gene's dominant non-NMD isoform onto the comparator's spliced sequence and, when that start codon was present, reading the ORF from it; where the reference start codon was absent, the highest-Kozak-scoring ORF was used. Kozak context (the −3 purine and +4 G determinants, scored as a position-weight log-odds) was used to adjudicate between competing start codons, in particular to compare the reference-derived start with the TransDecoder-predicted CDS. This procedure recovers occult PTCs that length-biased CDS predictors miss by selecting a downstream, in-frame start that bypasses the true (PTC-generating) initiator.

Reanalysis of Tan et al. (2025)

The recent study by Tan et al [13] also reports sharing of cell type-specific NMD effects across different cell types. To compare these estimates, cross-cell type sharing on a more equivalent footing between their study and ours, we reanalyzed the transcript-level EBSeq results from Tan et al. Supplementary Tables S1, S2, S4, and S6 (UPF2-KD, UPF2-iKO, UPF2-iKO&KD, and UPF3B-KO, each in hESCs and NPCs) with the same mashr framework used here, in place of the independent per-condition thresholding approach used in that study.

For each transcript and condition we set the effect size to $\hat{b}$ = $\log_2$(PostFC) and approximated its standard error as $\hat{s}$= $|\log_2(\text{PostFC})| / \Phi^{-1}(1 - \text{PPEE}/2)$, treating the EBSeq posterior probability of equivalent expression (PPEE = 1 − PPDE) as a two-sided significance proxy for weighting only; transcripts with PostFC ≤ 0 or = 1, PPEE ≥ 0.99, or non-positive $\hat{s}$ were excluded. Mashr was fit jointly across the eight conditions as described for our primary analyses (data-driven plus canonical covariances; null correlation from a random feature subset), yielding posterior mean $\log_2$ fold changes and local false sign rates. A transcript was called an NMD target in a condition if lfsr < 0.05 and posterior mean > 0; UPF2-dependent targets per cell type were the union of the three UPF2 conditions, and cross–cell type overlap was computed as the shared transcript count relative to each cell type's set (matching Tan et al.). For comparison we also reproduced the original threshold-based overlap using PostFC cutoffs (>2 and >1.5).

Isoform Pairs Analysis

In order to link splicing events to NMD susceptibility, we performed an analysis in which we focused only on annotated protein coding genes with at least three expressed isoforms (≥5% of overall gene expression in either DMSO or SMG1i and ≥5 reads in ≥1 sample) that met the following criteria: 1) the most highly expressed isoform in the gene was not susceptible to NMD, 2) the gene contained an expressed NMD susceptible isoform, 3) it contained at least one other expressed non-NMD isoform. For these genes, we then analyzed these three isoforms so that we could identify the splicing events that differentiated isoforms 2) and 3) from the dominant isoform, using this as a proxy for the splicing events that might have been responsible for the creation of those isoforms. This design ensures that when we compare NMD-related splicing transitions to Control splicing transitions, these comparisons share an identical reference isoform. To ensure the reference isoform represented the gene's dominant transcript rather than a minor non-NMD isoform, we further required it to account for at least 25% of the gene's total isoform expression in DMSO (averaged across the four cell types), yielding 1,548 gene-matched triplets. We developed a suite of analytic functions contained in a package called Isopair which is publicly available on Github (https://github.com/peter4244/Isopair). Non-NMD susceptible isoforms were defined as those with mashr lfsr ≥ 0.05 and per-cell-type adj.P.Val > 0.30 in the limma-voom step underlying mashr.

Within each pair, Isopair detects the splicing events that distinguish the comparator from its reference. Twelve event categories are enumerated (skipped exon, 5′ alternative splice site, 3′ alternative splice site, intron retention, intron-retention 5′-difference, intron-retention 3′-difference, intron-retention 5′-and-3′-difference, partial intron retention 5′, partial intron retention 3′, alternative transcription start site, alternative transcription end site, and missing internal exon block), and correctness is verified by the ability to achieve perfect reconstruction of the reference exon structure from the comparator.

For PTC analysis, comparator stop-codon position is intersected with splice junction structure to determine PTC status under the canonical 50-nt rule (a stop codon is a PTC if it falls >50 nt upstream of the last exon-exon junction, producing one or more downstream EJCs after splicing). Three nested cohort scopes are used downstream: a strict all-annotated scope in which the three isoforms of the triplet are all GENCODE-annotated coding transcripts (n = 130); a broader reference AUG-traceable scope in which the reference is annotated and its start codon can be projected onto each comparator's spliced sequence, re-intersected on (gene, reference) so the NMD and Control arms share the same gene set (n = 819); and an occult-PTC subset of the reference AUG-traceable scope in which the standard CDS caller does not flag a premature stop that the reference-anchored analysis reveals (n = 348). Because most NMD susceptible isoforms are novel, CDS regions are taken first from SQANTI3/TransDecoder (TD2) predictions and then, for isoforms classified as PTC-negative, recomputed from the reference isoform's annotated start codon ("reference AUG-tracing") to detect occult PTCs introduced by splicing-induced frameshifts. Splice events are then attributed as causal for PTC introduction by tracing which splicing event introduces or shifts the stop codon or terminal EJC. PTC-causing events are further classified into three mutually exclusive mechanism categories — frameshift (length change not a multiple of three nucleotides), in-frame stop (the event preserves the reading frame but introduces a new in-frame stop), and 3′ UTR splice (the event removes sequence between the natural stop and the terminal exon-exon junction).

Deep Learning Model

We trained an attention-based deep learning model to predict NMD susceptibility from transcript sequence and ORF-level structural features. Data were split into train/validation/test sets reserving chromosomes 2 and 4 for validation, chromosomes 1, 3, 5, and 7 for test, and the remaining autosomes plus sex chromosomes for training. Paralogous genes excluded from the test set.

Each transcript is represented as up to K=5 candidate ORFs, selected by priority: (1) the reference CDS ORF when the gene's dominant non-NMD isoform's start codon is present in the target isoform; (2) the SQANTI3/TransDecoder2 (TD2) CDS ORF when different from the reference; (3) the remaining slots filled by the top Kozak-scoring ORFs from a Kozak-aware ORFik[14] scan. This priority scheme allows the model to choose between likely ORFs when determining NMD susceptibility.

For each of the K=5 ORFs, the model receives two 500-nt sequence windows (centered on the middle nucleotide of the AUG and stop codon, respectively) encoded with 9 channels per position - A/C/G/T one-hot, exon-exon junction position, rolling 50-bp GC fraction, and three reading-frame one-hot encoded channels relative to the ORF's AUG - together with five per-ORF structural features. These structural features are: location of ORF start as a proportion of total transcript length (frac_start where an AUG at the 5' most position would have a value of 0), the same type of encoding for stop position (frac_stop), binary indicator for whether ORF is the same as the reference isoform (is_ref_cds), binary indicator for whether ORF was the TD2 called ORF (is_sqanti_cds), and the number of splice junctions located downstream of the stop codon (n_downstream_ejc). Two shared-weight CNN branches encode the AUG and stop windows; a third linear branch encodes the structural features; the three sub-embeddings are fused into a per-ORF embedding and aggregated across the 5 ORFs by an attention layer whose weights are interpretable as ribosome ORF selection probabilities. A small classification head maps the transcript embedding to the NMD prediction. The model was trained using binary cross-entropy (BCEWithLogitsLoss) with a pos_weight derived from training-set class frequencies (n_neg / n_pos) to account for class imbalance. Optimization was performed via the Adam optimizer (initial learning rate 1e-3) with differential weight decay (1e-3 for the AUG and stop window CNN parameters; 1e-4 for all other parameters), a batch size of 256, and PyTorch automatic mixed precision (fp16). The learning rate was dynamically reduced upon reaching plateaus in validation AUC (ReduceLROnPlateau, factor 0.5, patience 5), and training was terminated using early stopping based on the same metric (patience 10) for a maximum of 100 epochs, using a fixed random seed of 42. We swept twelve combinations of AUG window size ({100, 500, 1000} positions) × stop window size ({100, 500, 1000, 2000} positions) and selected the AUG=500 / stop=500 configuration by held-out AUC.

For interpretability we used three complementary attribution methods. Joint DeepSHAP[15] was used to evaluate the relative contribution of the start and stop sequence windows along with the matrices of ORF-related features. This procedure varied the AUG window + stop window + 5 structural features associated with the ORF-0 (most likely ORF) simultaneously while holding ranks 1–4 fixed at observed values, yielding per-position per-channel sequence attributions and per-feature structural attributions for every test transcript. We ran 5 independent replicates with different random seeds (500 background transcripts each) and averaged the resulting attributions. To evaluate the importance of individual features within these three main groups of features, KernelSHAP at the fusion layer computed the exact additive Shapley decomposition of the prediction across the three sub-encoders (AUG branch, stop branch, structural branch). Subgroup-specific attributions were computed by stratifying the pooled DeepSHAP outputs by the mechanistic NMD subgroups defined in the results section. For occult-PTC validation we additionally enumerated ORFs with the ORFik package via Isopair::enumerateOrfs(), which performs an exhaustive Kozak-aware ORF scan independent of splicing context. ORF agreement between TD2, ORFik, and the reference AUG was assessed by start- and stop-codon coordinate match. Candidate ORFs were classified as credible upstream open reading frames if they simultaneously satisfied three structural criteria: open-reading-frame length below 200 nucleotides, both start and stop codons within the 5′ half of the transcript, and a Kozak position-weight-matrix score ≥ 1. Per-isoform attention weights from the aggregator layer were extracted to characterize the model's ORF-selection behavior, and within-class stop-codon frequencies at the priority-ORF stop codon were tabulated directly from the test set.

To benchmark our model against previously developed variant-level NMD predictors, we ran a head-to-head comparison against NMDetective-B[16] and the NMDEP-re-thresholded rule baseline[17] on the $n = 819$ reference AUG-traceable cohort (756 NMD+/PTC+ + 63 NMD+/PTC− + 819 matched Controls; 1,638 isoforms total). The continuous gold standard for every isoform was the mashr posterior-mean log fold-change under SMG1 inhibition, aggregated across the four manuscript cell types (AT2, LAE, FB, MV) by taking the maximum per-isoform value. For each isoform we computed the four NMDetective-B features — in-last-exon membership of the (reference-AUG-projected, for PTC+/PTC−; natural, for Controls) stop codon, transcript-coordinate distance from start to stop, length of the exon containing the stop, and within-50-nt-of-last-EJ flag — directly from the long-read-observed transcript structure rather than by canonical-transcript lookup. NMDetective-B was scored with the published thresholds 50 / 150 / 407 nt and the leaf-level NMD-efficacy values (last-exon 0.00, start-proximal 0.12, long-exon 0.41, 50nt-rule 0.20, trigger 0.65 from Figure 1c of Lindeboom et al.); the NMDEP rule baseline used the same decision tree with NMDEP's grid-search-optimized thresholds 49 / 120 / 355 nt (Table 3 from Saadat et al.) and identical leaf values. The head-to-head intersection of isoforms scored by all three models was 1,570 (735

PTC+ + 54 PTC− + 781 Controls). For each model × stratum (pooled, NMD+/PTC+, NMD+/PTC−, Control) we computed Spearman correlation, Pearson correlation, R^2, mean absolute error, and root mean squared error against the gold-standard log fold-change. Because our model's predictions include training-set isoforms, all within-subclass metrics were additionally recomputed on the chr1 / chr3 / chr5 / chr7 held-out test split alone (n = 415; 195 PTC+ + 16 PTC− + 204 Controls) as a sensitivity analysis to confirm that the qualitative patterns were not artifacts of train-set inflation. Full-rank NMDetective-A and the full NMDEP MLP were deferred from this comparison because the full models are not publicly available.

Supplemental Results

Tan et. al. Reanalysis

The sharing of NMD effects we observed across lung cell types contrasts with a recent report by Tan et al.[13] that concluded that NMD susceptible mRNAs are predominantly cell type-specific, finding that only 16-25% of UPF2-dependent NMD susceptible transcripts were shared between human embryonic stem cells (hESCs) and neural progenitor cells (NPCs) (176 shared of 709 hESC and 1,088 NPC targets). We reasoned that the discrepancy could arise from how targets are compared across conditions: Tan et al. defined NMD targets by applying independent per-condition significance and fold-change thresholds in each cell type, a different approach than our mashr approach that explicitly models sharing between cell types. Accordingly, we reanalyzed the Tan et al. RNA-seq dataset using the same mashr-based approach used throughout this study (lfsr < 0.05 and positive posterior mean). Under this analysis, the transcript-level overlap of UPF2-dependent NMD targets between hESCs and NPCs increased to 41–50% (3,069 shared transcripts; 49.8% of hESC and 40.6% of NPC targets) with concordant effect direction observed across the two cell types for the large majority of shared targets. Thus, mashr-based reanalysis of the same data reveals substantially greater cross-context sharing of NMD targets and general agreement between studies.

Cell Type Specific Pathway Enrichment:

Beyond the shared core, each cell type harbored a set of cell type-specific enriched pathways **(ST1; ST2)**. LAE harbored by far the largest cell type-specific program, spanning cilium movement and cilium- or flagellum-dependent cell motility, epithelial differentiation (keratinization and cornified-envelope formation), glycolysis, the HSF1-mediated heat-shock response, and nucleocytoplasmic transport. AT2-specific pathways centered on translation-regulatory programs and lipid and fatty-acid metabolism (fatty-acid oxidation, acyl-CoA and arachidonate metabolism, and protein–lipid complex assembly). FB-specific pathways centered on energy metabolism (glucose, ketone, and selenoamino-acid metabolism, mitochondrial β-oxidation) and genome maintenance (telomere maintenance and recombinational repair). MV-specific pathways centered on amino-acid and glutamate transmembrane transport, mitochondrial protein import and inner-membrane organization, and endothelial functions including leukocyte transendothelial migration and collagen biosynthesis. Thus while the core NMD regulatory program is conserved across cell types, NMD additionally regulates certain pathways in a cell type-dependent manner.

Permutation-based Tests of Productive Fraction Analysis

For the 17% of genes where we observed a significant change in the productive fraction, we applied a series of donor- and platform-based tests to confirm that these changes reflected reproducible signal rather than technical noise **(SF24)**. To ask whether the within-donor direction of change was more consistent than expected if the direction were random, we used a sign-flip permutation test, and we observed that the number of consistent effects far exceeded the permutation null in AT2, LAE, and MV (permutation $p \leq 10^{-3}$)[8]. To ask whether donors agreed on direction more often than chance, we compared cross-donor sign concordance to an exact binomial expectation, finding 1.2- to 2.6-fold more concordant genes than expected ($p \leq 10^{-3}$). Finally, to ask whether the variance of the observed effect-size distribution was wider than would be expected under the null, we compared the observed variance to the sign-flip permutation null (variance ratio 1.26–2.13). The signal was strongest in LAE, where Storey's π_0 estimator placed the fraction of genes carrying a real effect at 48%[9]. In FB, the signal was not robust across all of these evaluations, so we did not analyze those results further, but for the other three cell types all of the above evaluations suggested a clear signal of change in productive fraction for a subset of genes.

Predictors of Productive Fraction Change and Its Direction

To understand these significant changes in productive fraction, we asked two questions: 1) What determines whether a gene's productive output changes? 2) What determines the direction of that change? NMD susceptibility governed whether a gene responded at all. In one-vs-rest models it raised the odds of both an increase and a decrease by nearly identical amounts (odds ratios 1.85 and 1.89), marking a gene as responsive without setting the sign. For genes with a significant change in productive fraction, the direction of change was negatively correlated with the gene's level of baseline expression **(ST4)**. In other words, genes with high baseline expression level tended to show a decrease in productive fraction during SMG1i, and genes with low baseline expression levels showed an increase. Further, the decrease was concentrated in LAE, which was ~18-fold more likely than AT2 to lose productive output, consistent with it harboring the largest and most cell-type-specific NMD program. A cross-validated model predicted the sign of change from independent short-read gene logFC almost as well as from all features combined (AUC 0.93 vs 0.96), but barely from NMD status alone (0.54). NMD inhibition therefore imposes no direction-specific penalty, and productive output simply rises or falls with the gene's underlying transcription **(ST4)**.

Productive Response in Non-NMD Genes

We wanted to see if the productive response extended to genes without an NMD susceptible isoform which we term non-NMD genes. Running the same analysis, we found 36 (AT2) to 408 (LAE) downregulated non-NMD genes and 160 (MV) to 252 (LAE) upregulated non-NMD genes. The NF-κB/TNFα inflammatory up-arm and the DNA repair down-arm were also present in non-NMD genes, indicating a cell-wide response rather than one confined to direct NMD targets. These genes have no NMD susceptible isoform, so their productive change must be a secondary, downstream consequence of NMD inhibition which is supported by the smaller number of significant genes.

Attributing NMD Susceptibility to Specific Splicing Events via Isopair Analysis

Isopair identifies 12 different categories of splice events (see **SF25 - Isopair Splice Events**), and correctness is verified by the ability to reconstruct the reference isoform completely given only the comparator isoform and the splice event list. The analysis package is available at https://github.com/peter4244/Isopair. The flow chart documenting the construction of these isoform NMD and control pair sets is shown in (**SF26 - Isopair Analysis Data Flow**). In the primary Isopair analysis set (n=1548 NMD susceptible and Control pairs), these genes had a median of 7 isoforms each (**SF27 - Isoform Count in Isoform Pair Sets**), and the median reference isoform accounted for 64% of its parent gene expression, with 68% of the reference isoforms accounting for >50% of parent gene expression (**SF28 - Reference Isoform Share of Gene Expression**). We then identify the "reference isoform", i.e. the highest expressed non-NMD isoform, against which we can compare both the NMD susceptible isoform (the NMD comparison) and the next highest expressed non-NMD triggering isoform (the Control comparison). As expected, the NMD and non-NMD comparator isoforms were expressed at lower levels than the reference isoform, but both comparator isoforms were expressed at similar levels in the SMG1i samples. Transcript length was similar for reference and NMD comparator isoforms (median length 2896 and 3018 nucleotides, respectively), and Control comparator isoforms were slightly shorter (median 2742 nucleotides, $p<0.001$ for comparison to both reference and NMD comparator, **SF29 - Transcript Length Comparison**).

When comparing NMD pairs to Control pairs, we tracked the balance of sequence gain versus loss, and we observed that terminal events (Alt TSS and TES) and skipped exons more frequently led to sequence gain in NMD isoforms, whereas intron retention events led to more sequence gain in controls (**SF30, Gain Direction by Event Type**). While there was a strong dose-response relationship between upstream distance of the stop codon and the magnitude of the NMD response up to ~350nt (**SF31 - PTC distance dose response**), the number of downstream EJCs showed little additional effect beyond the first EJC (**SF32 - NMD Effect Size by Number of EJCs**).

TD2 Bias Against PTC-Containing CDSs

In the Reference AUG-traceable set, we compared the TD2-called CDS to the CDS obtained from "reference AUG-tracing", i.e. defining the CDS by assuming that the comparator isoforms uses the same start codon as its paired reference. When we compared the two ways of calling the CDS, the reference AUG-traced and TD2-called CDS used the same start codon in 52% of cases. When start codons differed, the reference start codon usually had a higher Kozak score than the TD2-called start codon and reference CDS was shorter than the TD2 CDS (**SF34**), a trend that was much stronger in the NMD+/PTC- isoforms.

We identified 348 NMD susceptible isoforms where the reference AUG-traced CDS contained a PTC but the TD2-called CDS did not. Of those isoforms, 99% (345/348) of the

TD2-called CDSs were downstream from the reference AUG, demonstrating that when choosing a CDS in these cases TD2 skipped over the PTC-containing reference ORF in favor of a longer, non-PTC containing downstream ORF. To determine which of these ORFs had a stronger start codon, we calculated Kozak scores for the TD2 selected AUG and the reference AUG, and we observed that the reference AUG was stronger in 82% of cases (286/348, **SF35**). Collectively this demonstrates a negative bias for TD2 CDS calls against PTC-containing ORFs, so we calculated what the PTC rate in NMD+ isoforms would be if we relied on reference AUG tracing to determine the CDS.

3' UTR Length Analysis

Since there is some controversy about the importance of 3' UTR length as a driver of NMD in humans, we examined 3' UTR length in NMD+/PTC+, NMD+/PTC-, and Control isoforms in both the GENCODE restricted isoform set (n=130) and the more expansive pair set (n=819). There is another nuance in terms of calculating 3' UTR length, which is that in PTC+ isoforms 3' UTR length increases by virtue of the PTC, so we also calculated the 3' UTR length in these isoforms from the first stop codon available within 50nt of the terminal EJC. Using this estimated "natural" 3' UTR length, there was no significant difference in 3' UTR length between NMD+/PTC+ isoforms and the other groups in the GENCODE restricted set (**SF33**). In the expanded set, NMD+/PTC+ isoforms had significantly longer 3' UTRs than both other groups ($p<0.005$ - median length 1324/722/922 for NMD+/PTC+, NMD+/PTC-, and Control respectively) (**SF36**).

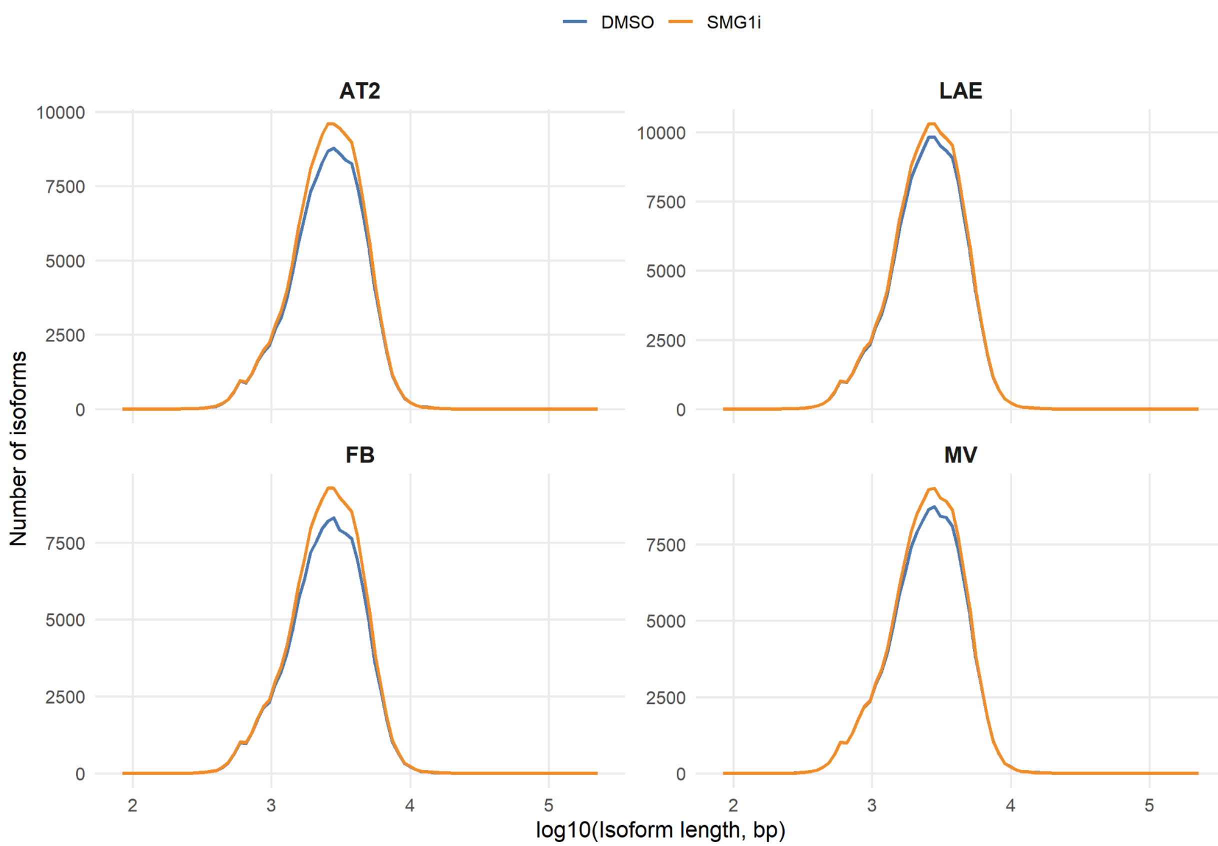
GC Content in Predictive Model Analysis

To prevent the GC signals from overwhelming more region-specific signals like Kozak sequence, the model incorporates a channel specifically for GC content, and it is clear that GC content is a strong signal differentiating NMD from Control isoforms in the STOP window after the stop codon (**SF42**). This likely reflects the effect of PTCs, which effectively turn exons (which are GC-rich relative to UTRs) into 3' UTR sequence.

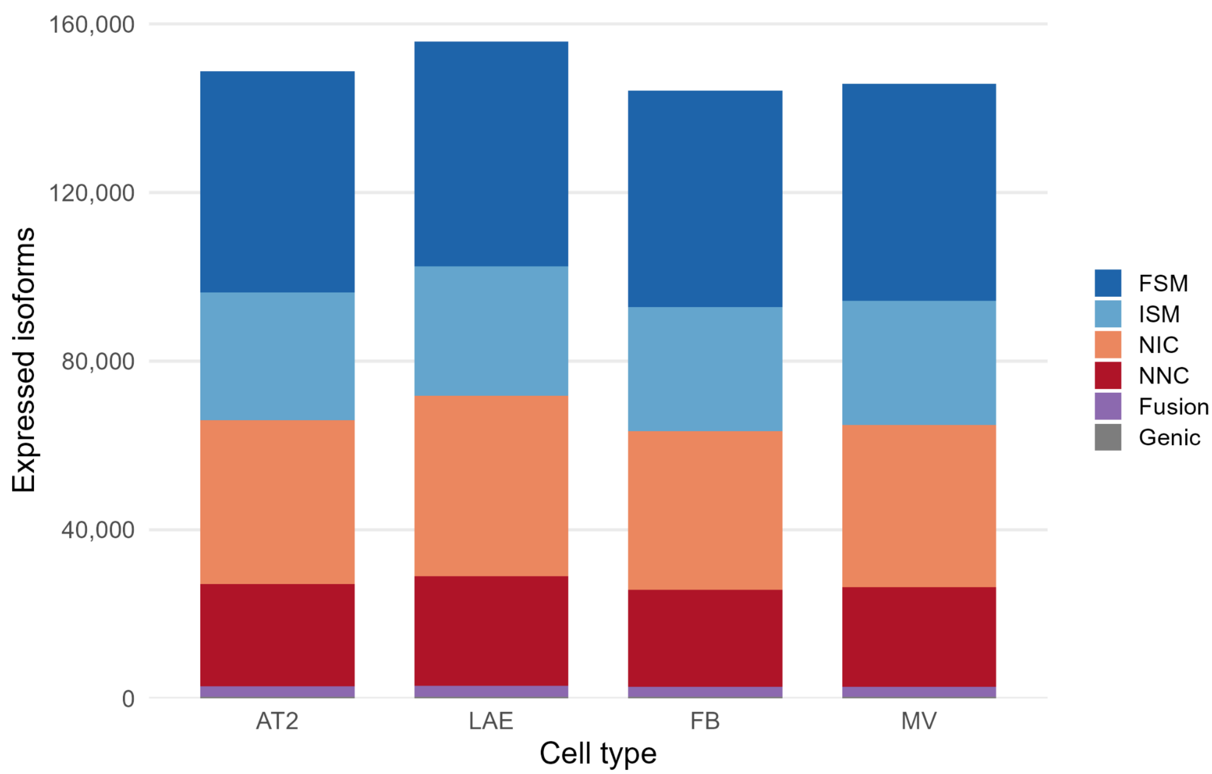
Sequence Based NMD Model Compared to Prior NMD Models

To benchmark our model's performance, we compared it to other predictive models for NMD - NMDetective-B and NMDEP, choosing for both the simpler rule-based models that could be replicated in our dataset. Performance was comparable for all three models, and our model, which had more continuous predictions had stronger correlation to the ground truth within NMD+/PTC+ and NMD+/PTC- classes (Supplemental Results and SF43). Focusing on isoforms from our gene-matched set in testing data only (n=415 isoforms from 216 genes), we compared predictions from each of the predictive models against the log2FC from the mashr analysis for NMD susceptibility, using the correlation between predicted and observed values from the test dataset as our evaluation metric. We observed that the three models had similar overall performance (Spearman rho = 0.77 for our model and 0.79 for NMDetective-B and NMDEP). When we stratified isoforms by NMD and PTC status (NMD+/PTC+, NMD+/PTC-, and Control), we observed higher correlations with our model predictions, indicating that our model seems to have learned additional features beyond the explicit rules in the other two models for making NMD predictions (**SF43**), though this also reflect the more gradated nature of our model predictions relative to the more discrete distribution of scores from the rule-based models.

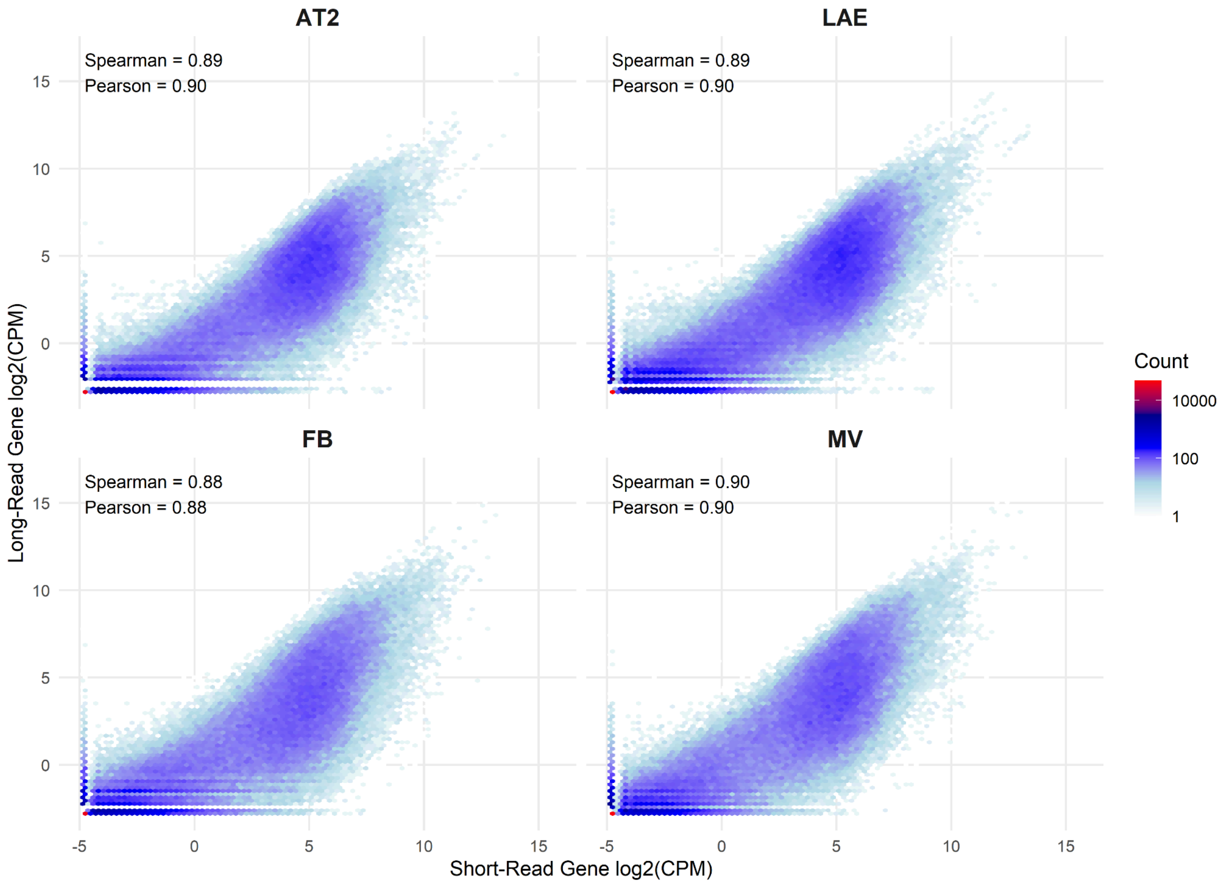
Supplemental Figures



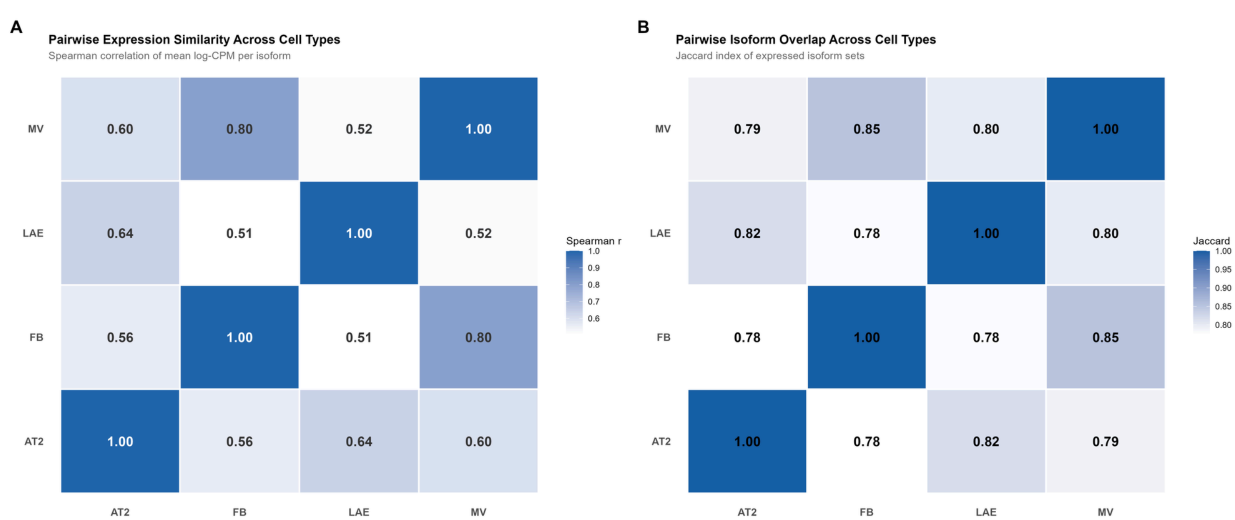
SF1 | Isoform Length by Cell Type and Treatment. Distribution of assembled long-read isoform lengths (log10 base pairs), shown as frequency curves for each of the four cell types (AT2, LAE, FB, MV) and overlaid by treatment (DMSO, SMG1i).



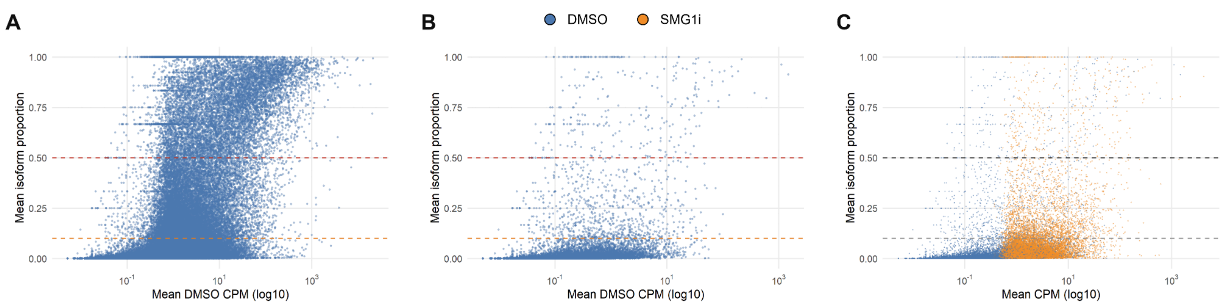
SF2 | SQANTI3 Structural Categories by Cell Type. Stacked bars of the number of expressed long-read isoforms in DMSO in each cell type (AT2, LAE, FB, MV), partitioned by SQANTI3 structural category (FSM, ISM, NIC, NNC, fusion, genic, antisense, intergenic, and genic-intron). **Abbreviations.** FSM, full splice match; ISM, incomplete splice match; NIC, novel in catalog; NNC, novel not in catalog.



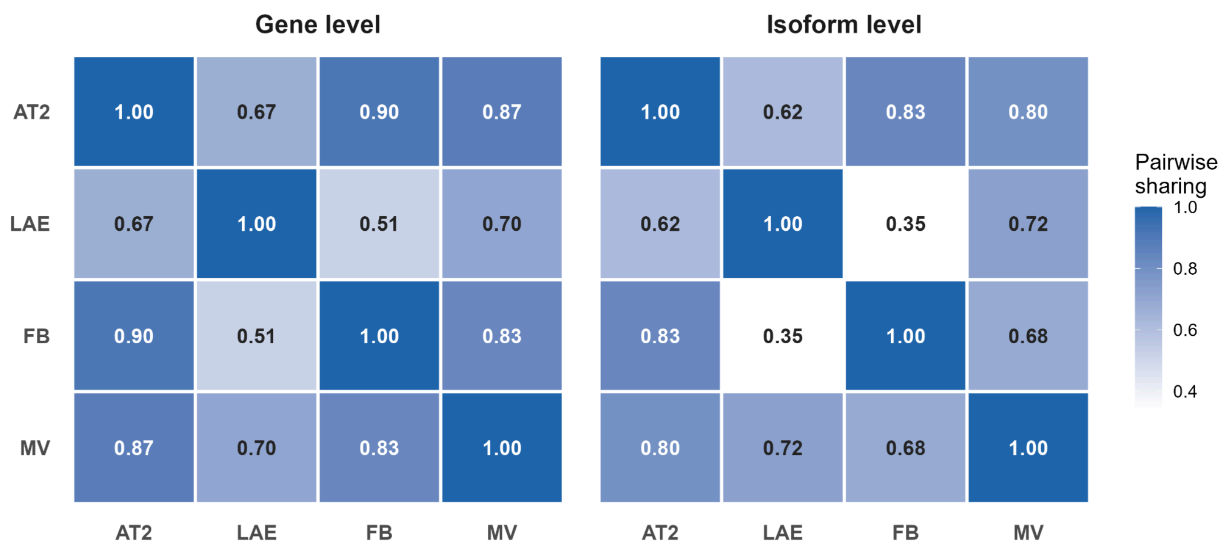
SF3 | Gene-Level Correlation of Short-read and Long-read Expression in All Cell Types. Hexbin density plots of short-read versus long-read gene expression (log2 CPM) for each cell type (AT2, LAE, FB, MV); color encodes bin count and the per-cell-type Spearman and Pearson correlations are annotated on each panel. **Abbreviations.** CPM, counts per million.



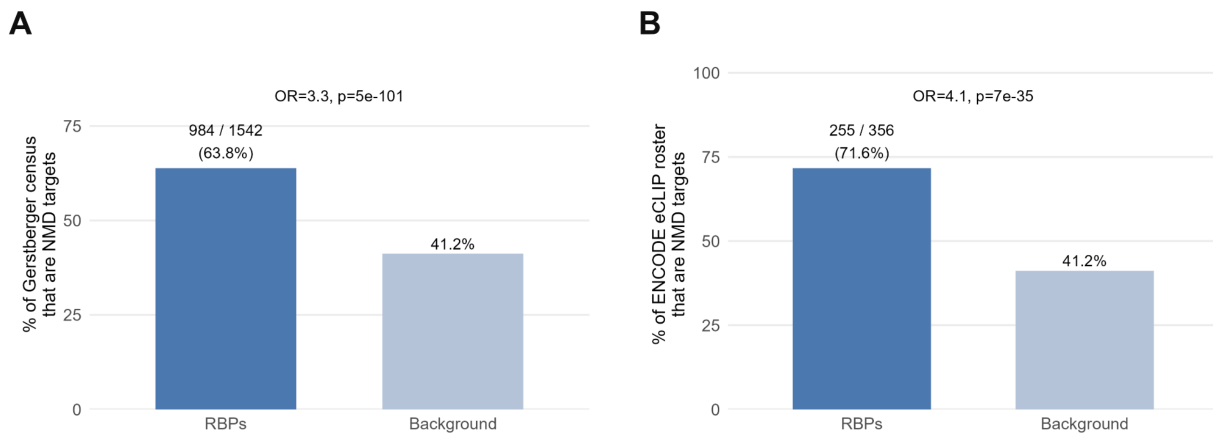
SF4 | Pairwise Expression Similarity and Isoform-Set Overlap Between Cell Types. A) Heatmap of pairwise Spearman correlations of mean per-isoform expression (log CPM) across the four cell types (AT2, LAE, FB, MV). **B)** Heatmap of pairwise Jaccard overlap of the expressed-isoform sets across the same cell types. **Abbreviations.** CPM, counts per million.



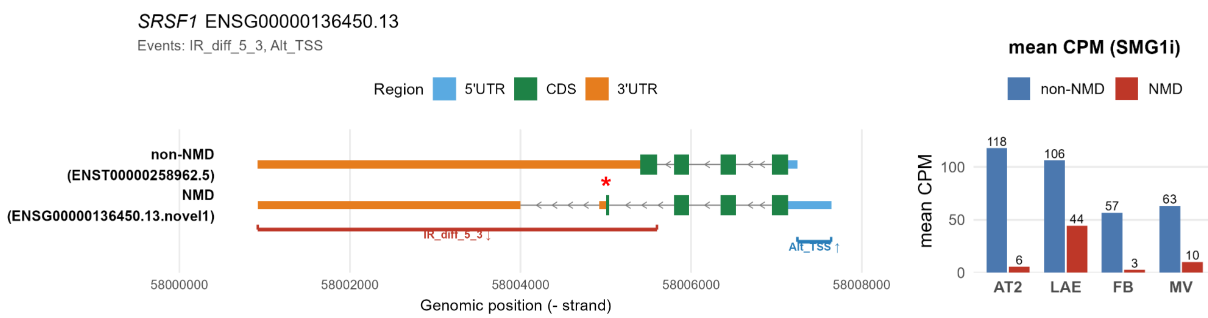
SF5 | Isoform Proportion versus Expression. A) All isoforms under DMSO and **B)** NMD susceptible isoforms, plotting each isoform's mean proportion of its gene's total expression against mean expression (log10 CPM), with reference lines at 0.10 and 0.50 proportion. **C)** NMD susceptible isoforms under DMSO versus SMG1i, with points colored by treatment. **Abbreviations.** CPM, counts per million.



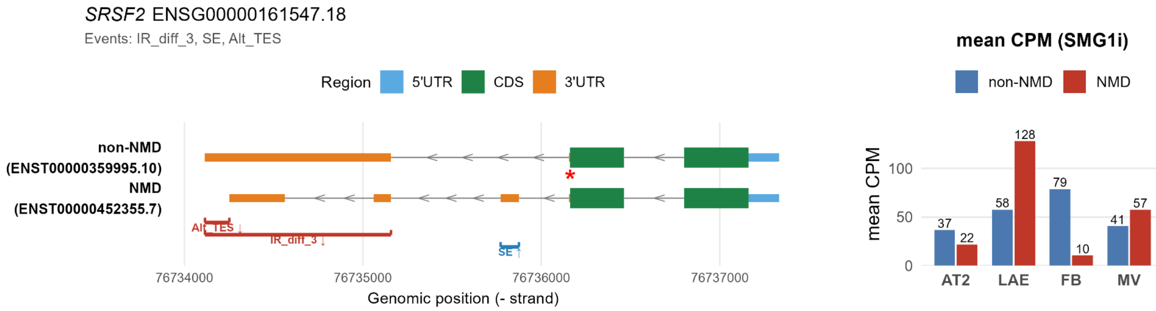
SF6 | Effect-Size Sharing at Gene and Isoform Resolution. Pairwise sharing heatmaps by mashr for short-read gene-level (left) and long-read isoform-level (right) NMD effects; each cell gives the fraction of features that share the same sign and lie within two logFC between the two cell types (AT2, LAE, FB, MV).



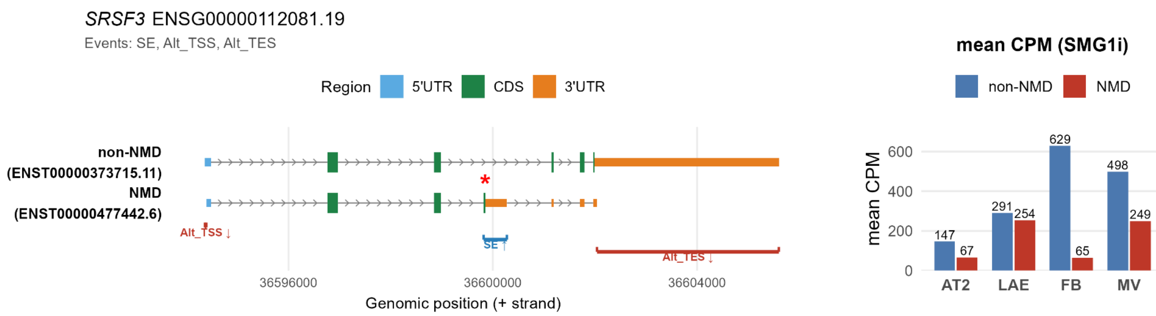
SF7 | RNA-Binding-Protein Enrichment among NMD Susceptible Genes. A) Fraction of the Gerstberger 2014 RBP census that are NMD targets versus the background rate. **B)** The same comparison against the ENCODE eCLIP roster (Van Nostrand 2020). An accompanying table reports enrichment for each RBP functional category (ST3). **Abbreviations.** RBP, RNA-binding protein; eCLIP, enhanced crosslinking immunoprecipitation.



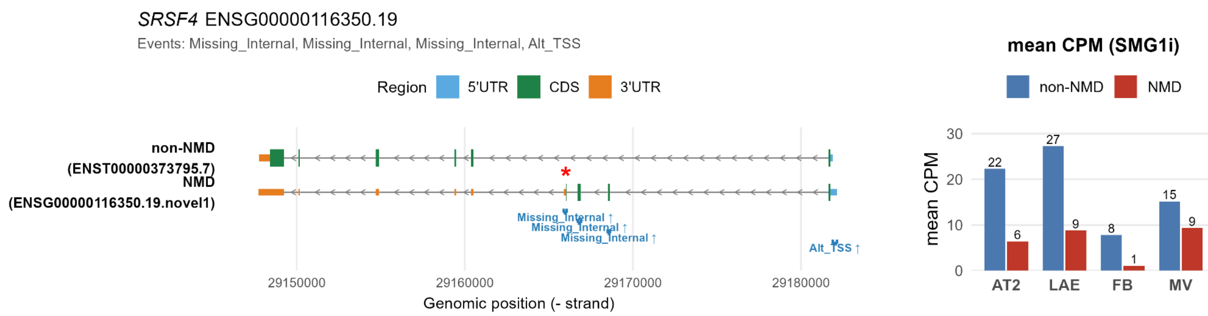
SF8 | *SRSF1* Isoform Structure and Expression (non-NMD vs NMD). Exon/CDS structure of the most-expressed non-NMD reference isoform in DMSO and the top expressed NMD susceptible isoform in SMG1i of *SRSF1*, with splicing events annotated and the premature termination codon marked by a star. The side panel shows each isoform's mean CPM under SMG1i across the four cell types (AT2, LAE, FB, MV). **Abbreviations.** PTC, premature termination codon; CPM, counts per million; IR_diff 5'3', intron retention with both 5' and 3' boundary mismatches; Alt TSS, alternative transcription start site.



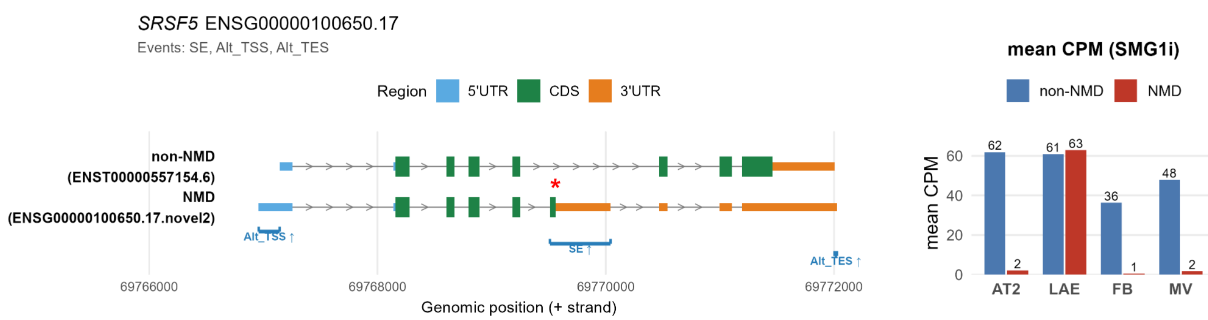
SF9 | *SRSF2* Isoform Structure and Expression (non-NMD vs NMD). Exon/CDS structure of the most-expressed non-NMD reference isoform in DMSO and the top expressed NMD susceptible isoform in SMG1i of *SRSF2*, with splicing events annotated and the premature termination codon marked by a star. The side panel shows each isoform's mean CPM under SMG1i across the four cell types (AT2, LAE, FB, MV). **Abbreviations.** PTC, premature termination codon; CPM, counts per million; IR_diff 3', intron retention with a 3' boundary mismatch; SE, skipped exon; Alt TES, alternative transcription end site.



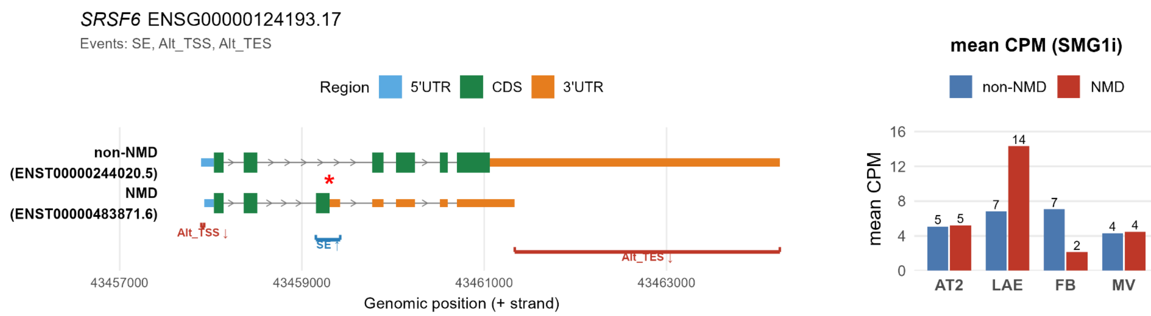
SF10 | *SRSF3* Isoform Structure and Expression (non-NMD vs NMD). Exon/CDS structure of the most-expressed non-NMD reference isoform in DMSO and the top expressed NMD susceptible isoform in SMG1i of *SRSF3*, with splicing events annotated and the premature termination codon marked by a star. The side panel shows each isoform's mean CPM under SMG1i across the four cell types (AT2, LAE, FB, MV). Abbreviations. PTC, premature termination codon; CPM, counts per million; SE, skipped exon; Alt TSS, alternative transcription start site; Alt TES, alternative transcription end site.



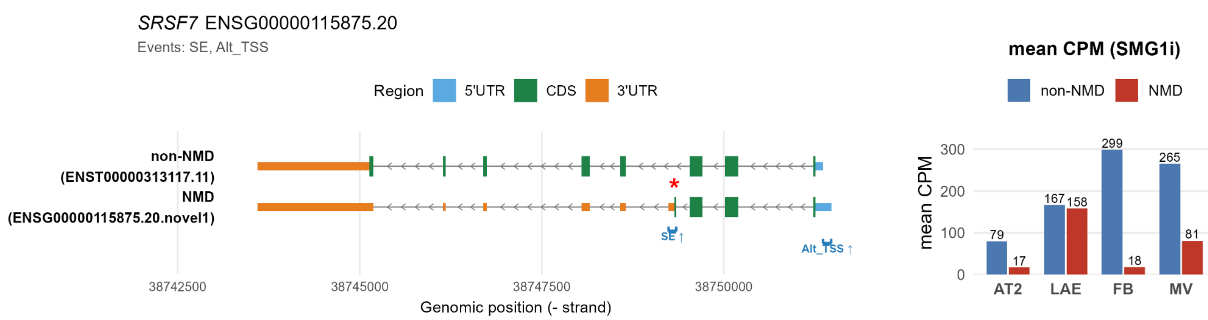
SF11 | *SRSF4* Isoform Structure and Expression (non-NMD vs NMD). Exon/CDS structure of the most-expressed non-NMD reference isoform in DMSO and the top expressed NMD susceptible isoform in SMG1i of *SRSF4*, with splicing events annotated and the premature termination codon marked by a star. The side panel shows each isoform's mean CPM under SMG1i across the four cell types (AT2, LAE, FB, MV). Abbreviations. PTC, premature termination codon; CPM, counts per million; Missing Internal, internal exon present in one isoform within the span of the other, outside strictly skipped exon criteria; Alt TSS, alternative transcription start site.



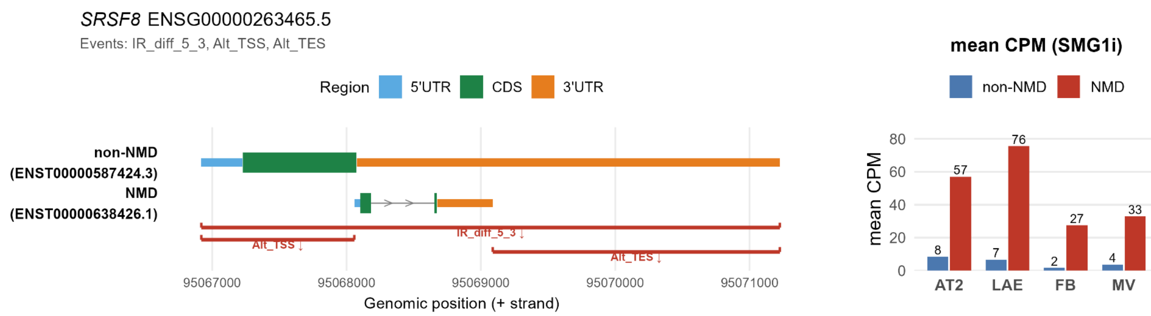
SF12 | *SRSF5* Isoform Structure and Expression (non-NMD vs NMD). Exon/CDS structure of the most-expressed non-NMD reference isoform in DMSO and the top expressed NMD susceptible isoform in SMG1i of *SRSF5*, with splicing events annotated and the premature termination codon marked by a star. The side panel shows each isoform's mean CPM under SMG1i across the four cell types (AT2, LAE, FB, MV). Abbreviations. PTC, premature termination codon; CPM, counts per million; SE, skipped exon; Alt TSS, alternative transcription start site; Alt TES, alternative transcription end site.



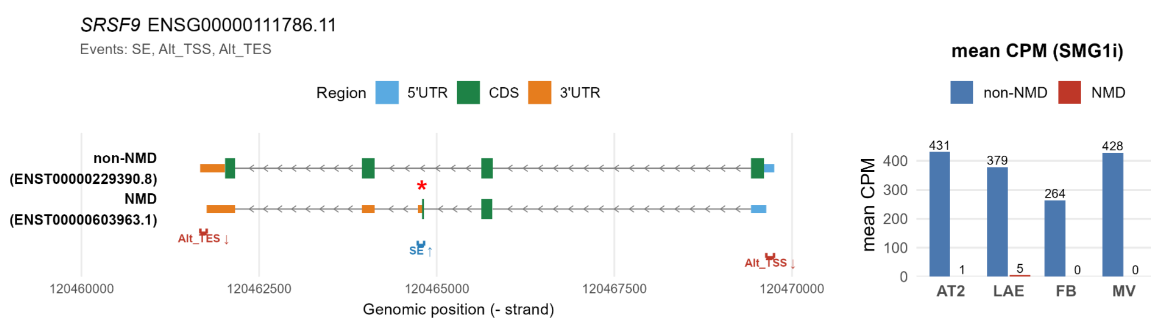
SF13 | *SRSF6* Isoform Structure and Expression (non-NMD vs NMD). Exon/CDS structure of the most-expressed non-NMD reference isoform in DMSO and the top expressed NMD susceptible isoform in SMG1i of *SRSF6*, with splicing events annotated and the premature termination codon marked by a star. The side panel shows each isoform's mean CPM under SMG1i across the four cell types (AT2, LAE, FB, MV). Abbreviations. PTC, premature termination codon; CPM, counts per million; SE, skipped exon; Alt TSS, alternative transcription start site; Alt TES, alternative transcription end site.



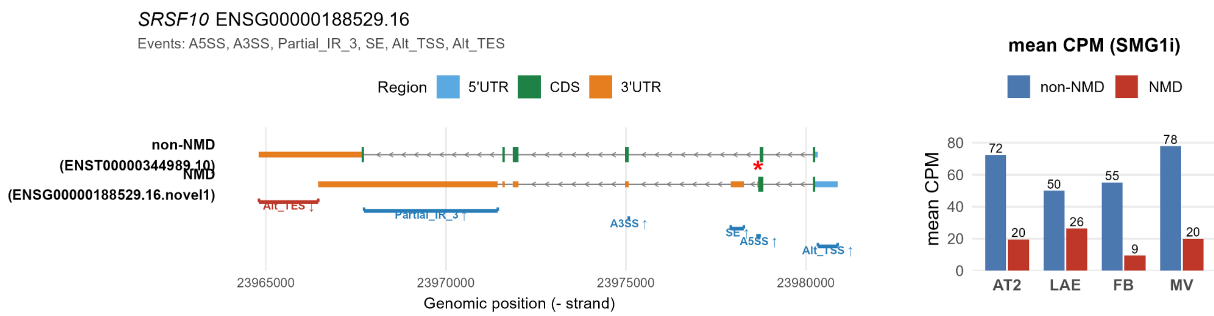
SF14 | *SRSF7* Isoform Structure and Expression (non-NMD vs NMD). Exon/CDS structure of the most-expressed non-NMD reference isoform in DMSO and the top expressed NMD susceptible isoform in SMG1i of *SRSF7*, with splicing events annotated and the premature termination codon marked by a star. The side panel shows each isoform's mean CPM under SMG1i across the four cell types (AT2, LAE, FB, MV). Abbreviations. PTC, premature termination codon; CPM, counts per million; SE, skipped exon; Alt TSS, alternative transcription start site.



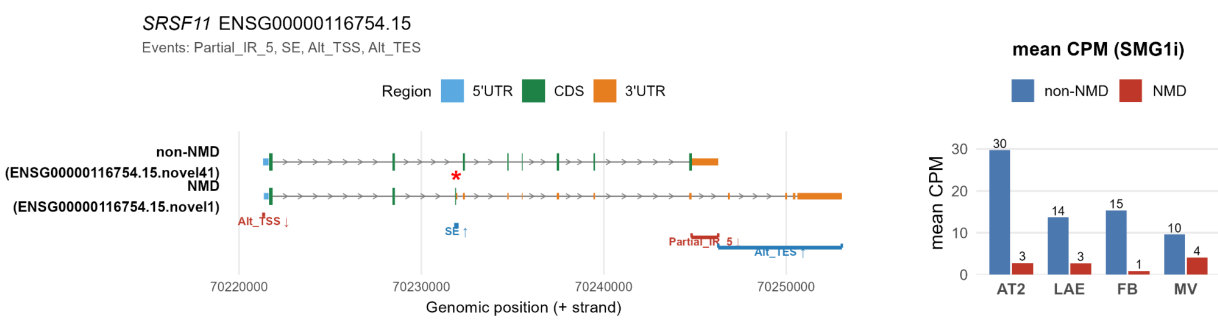
SF15 | *SRSF8* Isoform Structure and Expression (non-NMD vs NMD). Exon/CDS structure of the most-expressed non-NMD reference isoform in DMSO and the top expressed NMD susceptible isoform in SMG1i of *SRSF8*, with splicing events annotated and the premature termination codon marked by a star. The side panel shows each isoform's mean CPM under SMG1i across the four cell types (AT2, LAE, FB, MV). **Abbreviations.** PTC, premature termination codon; CPM, counts per million; IR_diff 5'3', intron retention with both 5' and 3' boundary mismatches; Alt TSS, alternative transcription start site; Alt TES, alternative transcription end site.



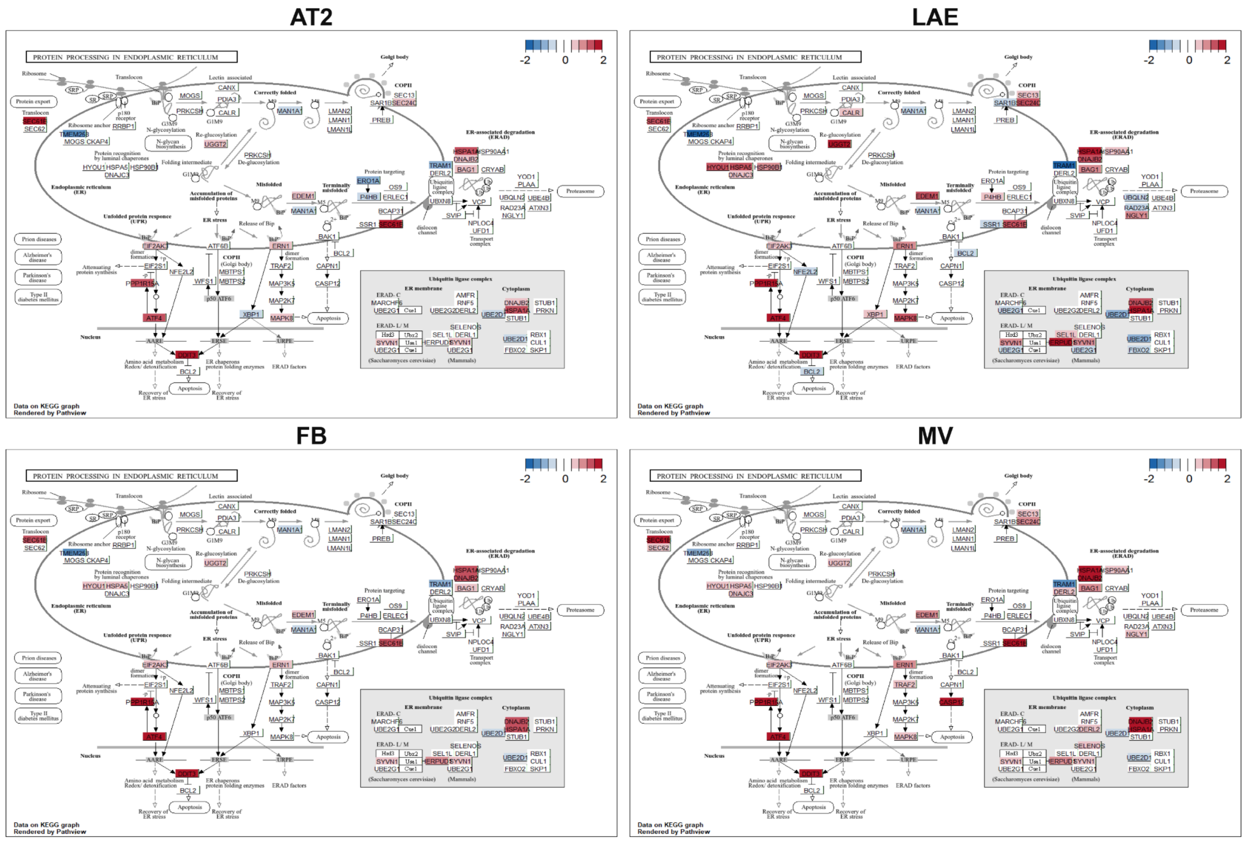
SF16 | *SRSF9* Isoform Structure and Expression (non-NMD vs NMD). Exon/CDS structure of the most-expressed non-NMD reference isoform in DMSO and the top expressed NMD susceptible isoform in SMG1i of *SRSF9*, with splicing events annotated and the premature termination codon marked by a star. The side panel shows each isoform's mean CPM under SMG1i across the four cell types (AT2, LAE, FB, MV). Abbreviations. PTC, premature termination codon; CPM, counts per million; SE, skipped exon; Alt TSS, alternative transcription start site; Alt TES, alternative transcription end site.



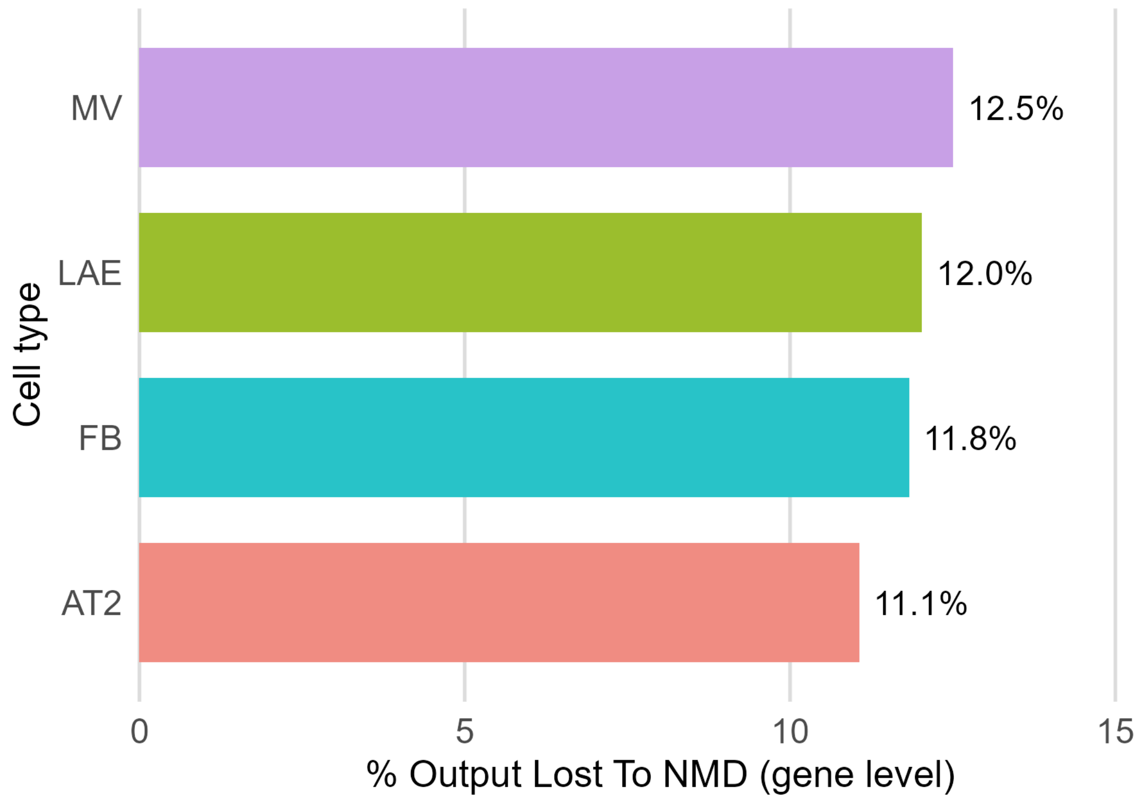
SF17 | *SRSF10* Isoform Structure and Expression (non-NMD vs NMD). Exon/CDS structure of the most-expressed non-NMD reference isoform in DMSO and the top expressed NMD susceptible isoform in SMG1i of *SRSF10*, with splicing events annotated and the premature termination codon marked by a star. The side panel shows each isoform's mean CPM under SMG1i across the four cell types (AT2, LAE, FB, MV). Abbreviations. PTC, premature termination codon; CPM, counts per million; A5SS, alternative 5′ splice site; A3SS, alternative 3′ splice site; Partial IR 3′, partial intron retention at the 3′ acceptor boundary; SE, skipped exon; Alt TSS, alternative transcription start site; Alt TES, alternative transcription end site.



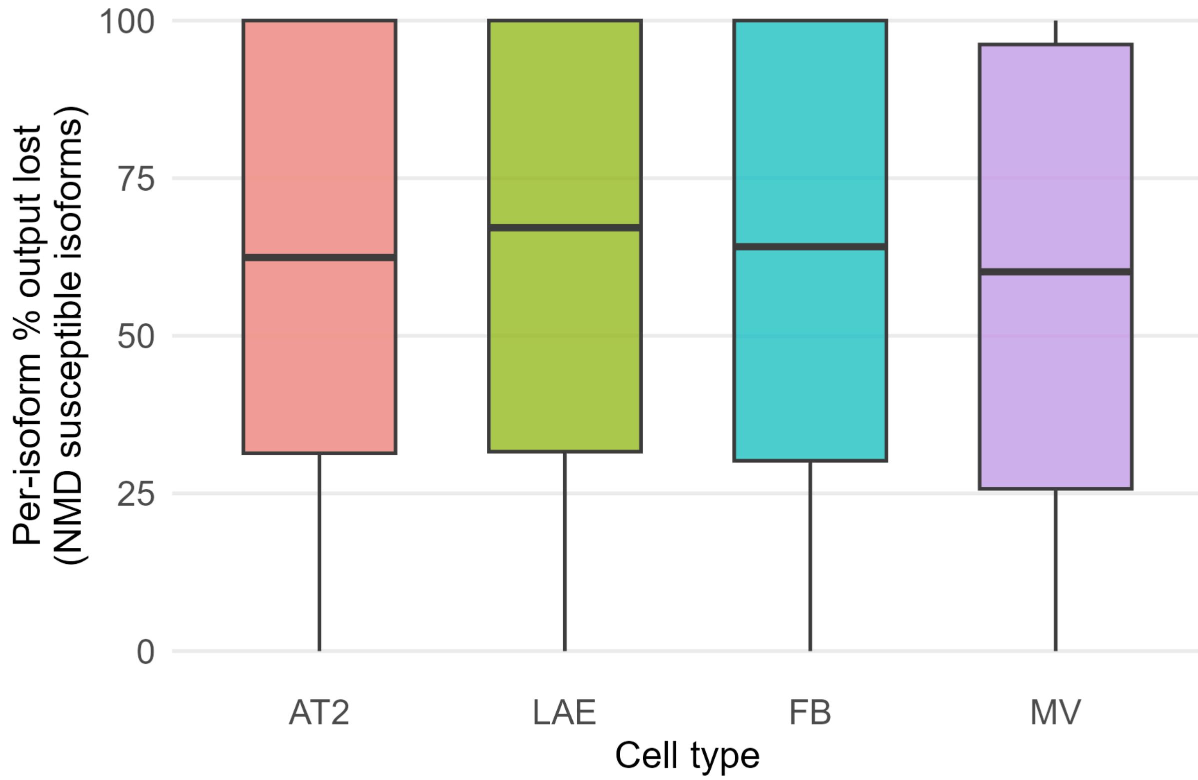
SF18 | *SRSF11* Isoform Structure and Expression (non-NMD vs NMD). Exon/CDS structure of the most-expressed non-NMD reference isoform in DMSO and the top expressed NMD susceptible isoform in SMG1i of *SRSF11*, with splicing events annotated and the premature termination codon marked by a star. The side panel shows each isoform's mean CPM under SMG1i across the four cell types (AT2, LAE, FB, MV). Abbreviations. PTC, premature termination codon; CPM, counts per million; Partial IR 5′, partial intron retention at the 5′ donor boundary; SE, skipped exon; Alt TSS, alternative transcription start site; Alt TES, alternative transcription end site.



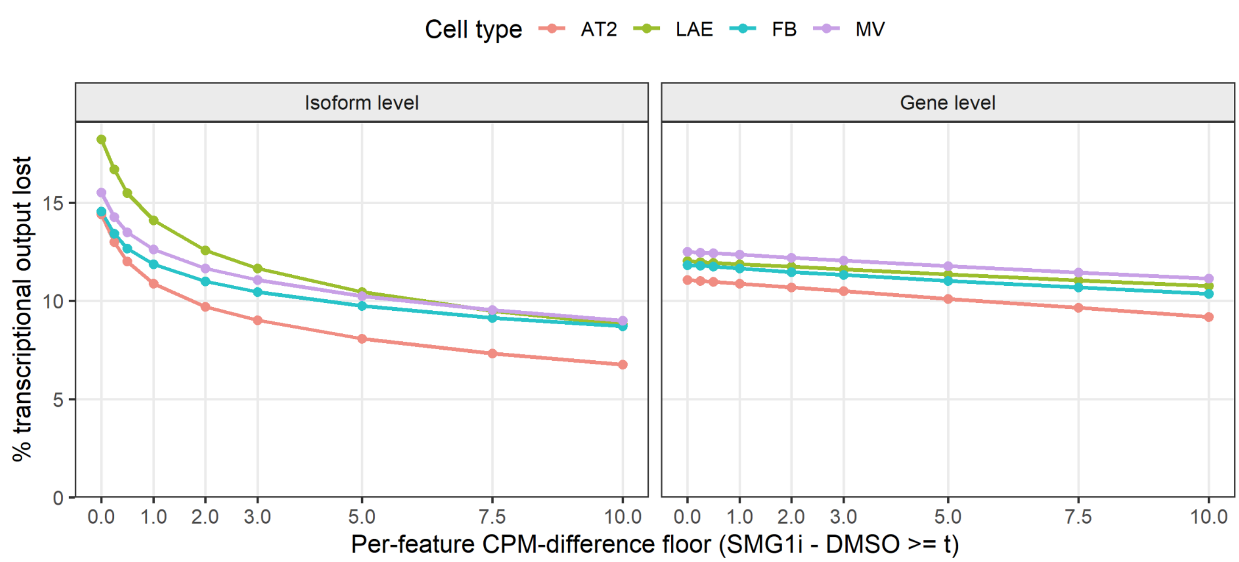
SF19 | Gene-Level NMD Susceptibility of the KEGG Protein Processing in Endoplasmic Reticulum Pathway in All Cell Types. The KEGG ER protein-processing pathway (hsa04141) shown for each cell type (AT2, LAE, FB, MV), with pathway genes colored by short-read NMD posterior log2 fold change. This pathway was selected because it contains ATF4, a key NMD-regulated transcription factor whose targets include multiple other pathway members, allowing visualization of how NMD susceptibility propagates both among co-regulated genes and to functionally downstream genes not themselves direct NMD substrates. Abbreviations. ER, endoplasmic reticulum; KEGG, Kyoto Encyclopedia of Genes and Genomes; NMD, nonsense-mediated decay.



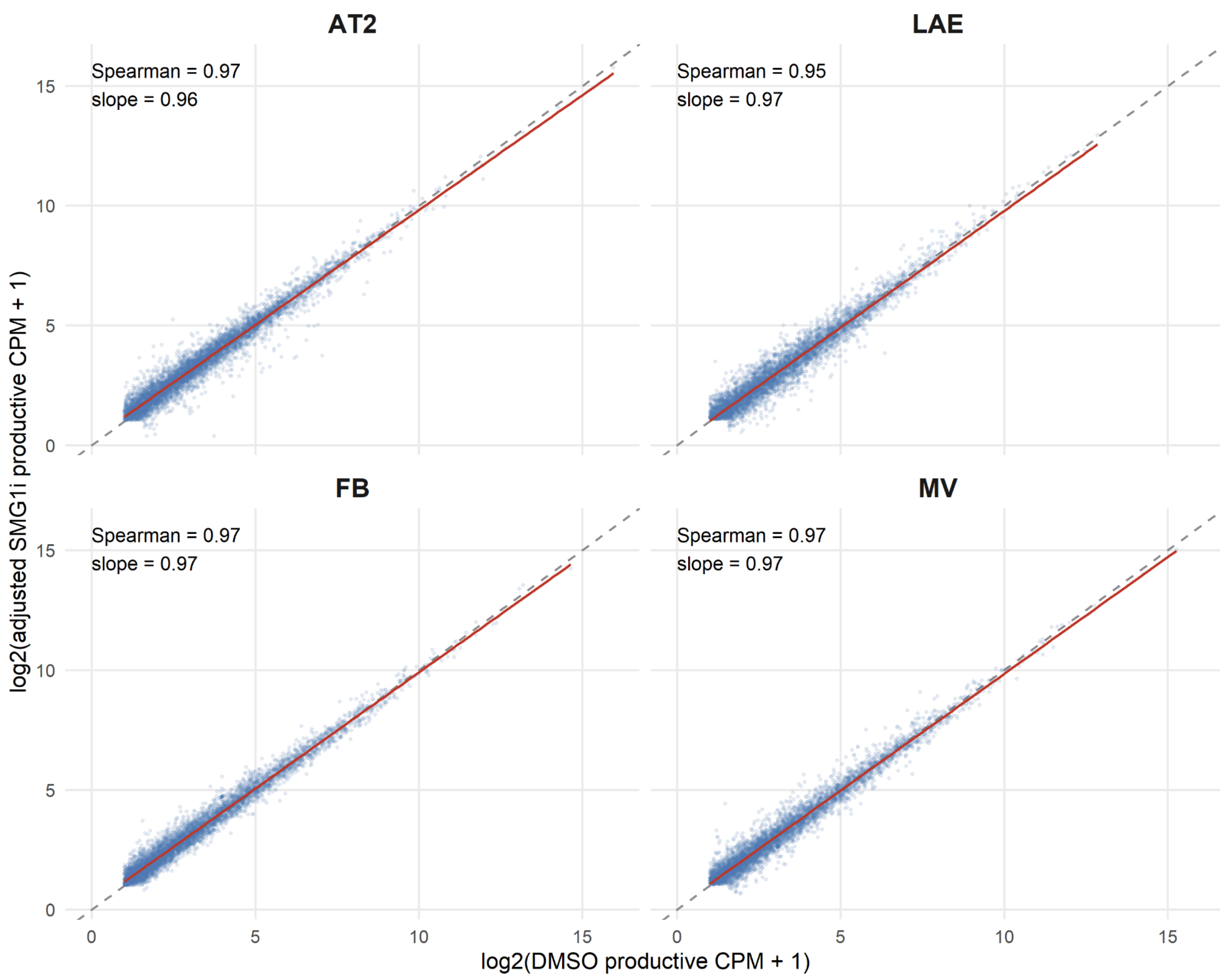
SF20 | Percent Transcriptional Output Lost to NMD, Gene Level. Per-cell-type percentage of gene-level transcriptional output attributable to NMD-susceptible isoforms (SMG1i minus DMSO), summarized across genes for each cell type (AT2, LAE, FB, MV).



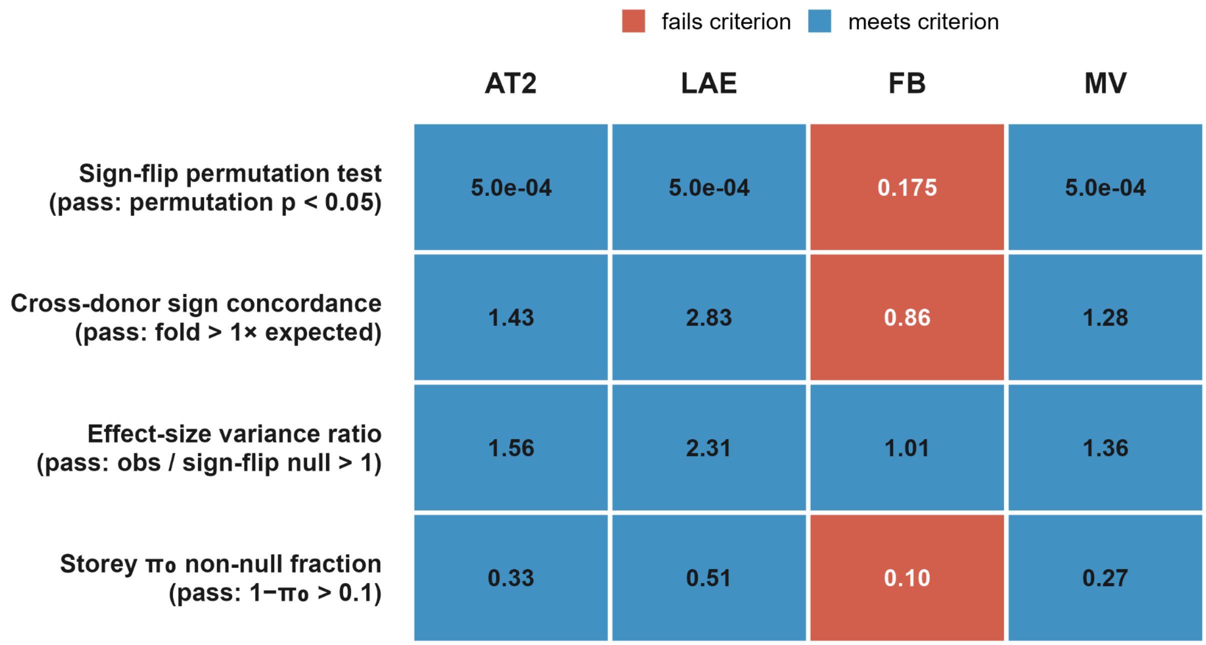
SF21 | Per-Isoform Percent Output Lost. Distribution of the percentage of a gene's total output contributed by each individual NMD-susceptible isoform, shown per cell type (AT2, LAE, FB, MV).



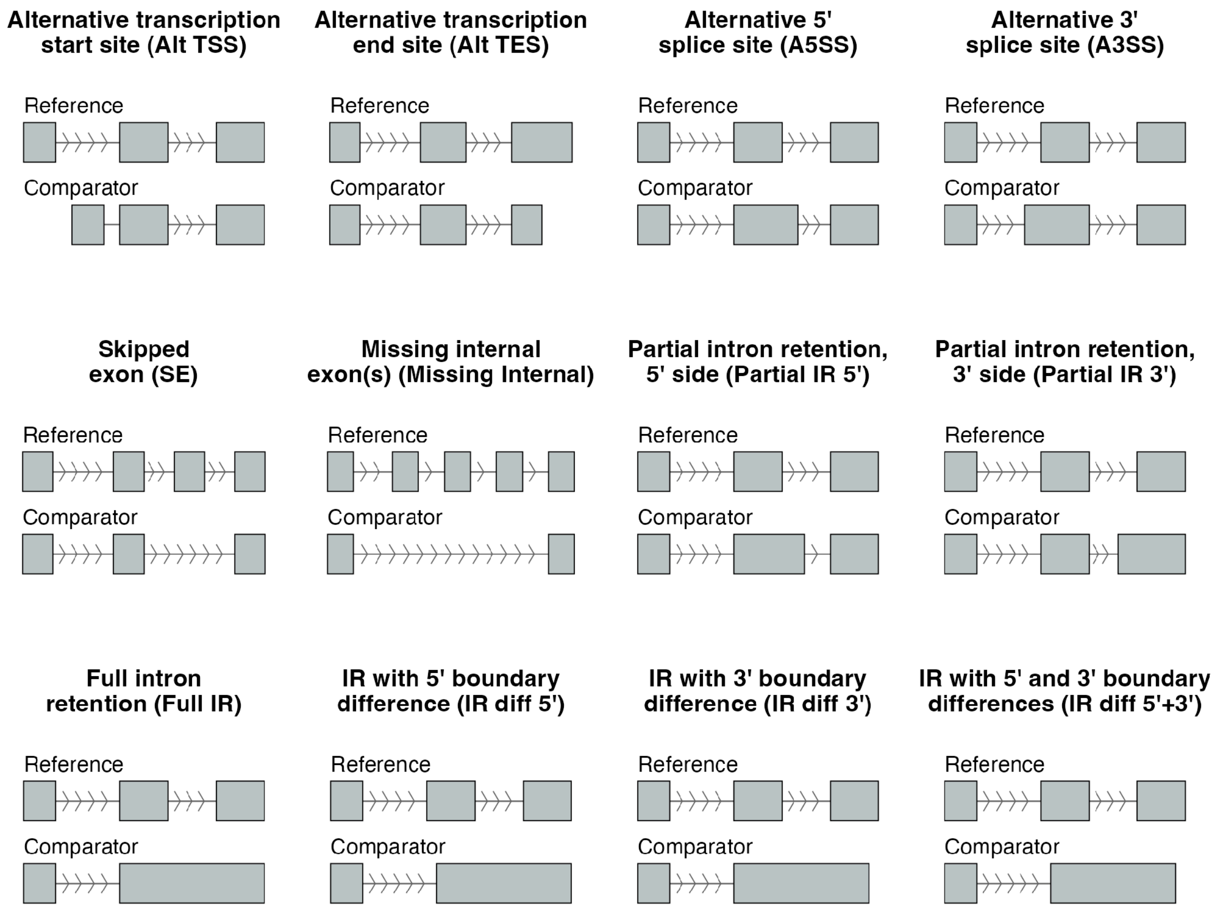
SF22 | Transcriptional Output Lost by CPM Thresholds. Percent Transcriptional Output lost recomputed with varying expression, in CPM, thresholds at the gene and isoform level for all cell types (AT2, LAE, FB, MV).



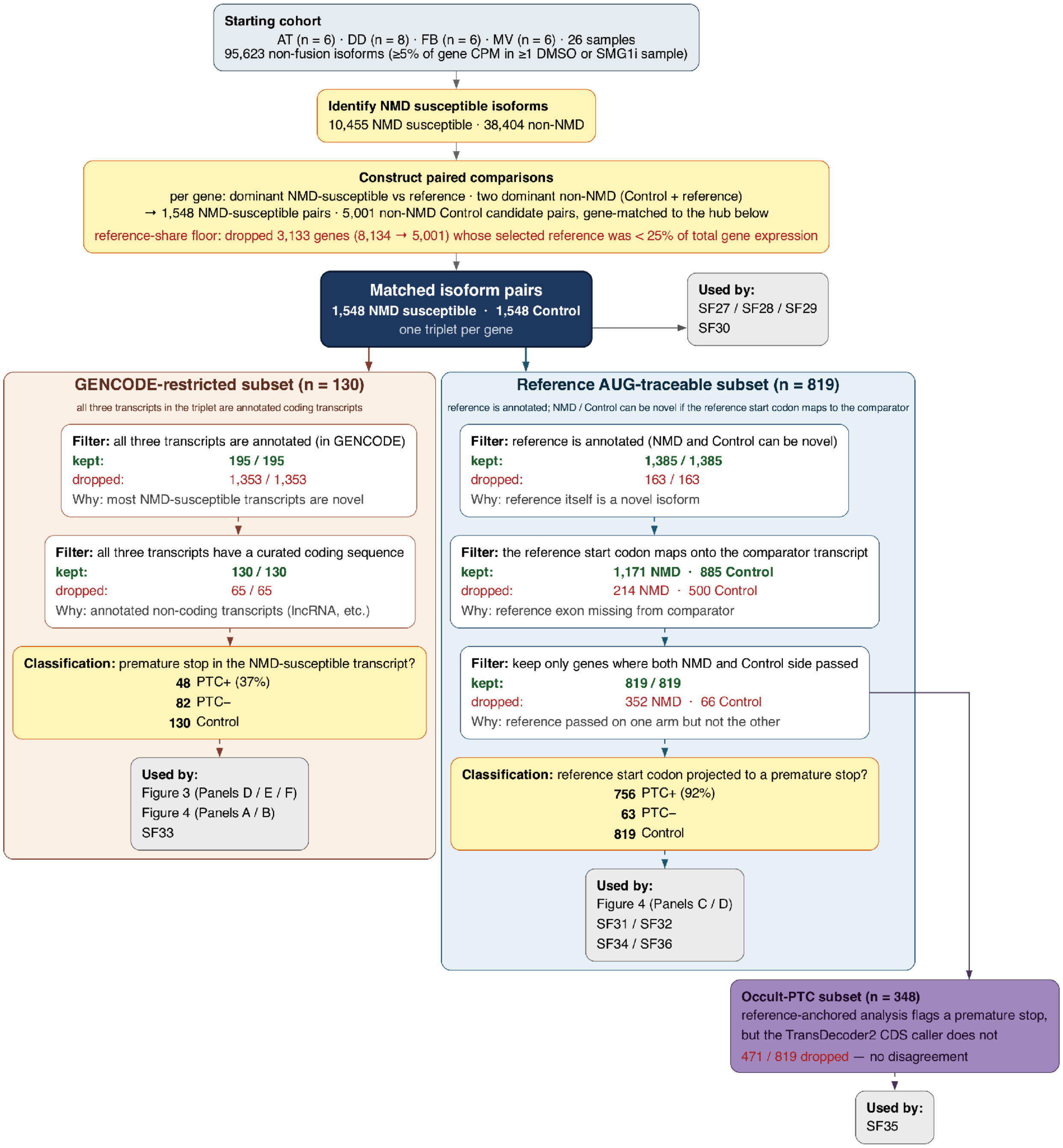
SF23 | Correlation Between non-NMD Gene Expression in DMSO and Adjusted SMG1i. Per-cell-type scatter of non-NMD gene expression under DMSO versus SMG1i after the custom NMD-anchored normalization, by cell type (AT2, LAE, FB, MV).



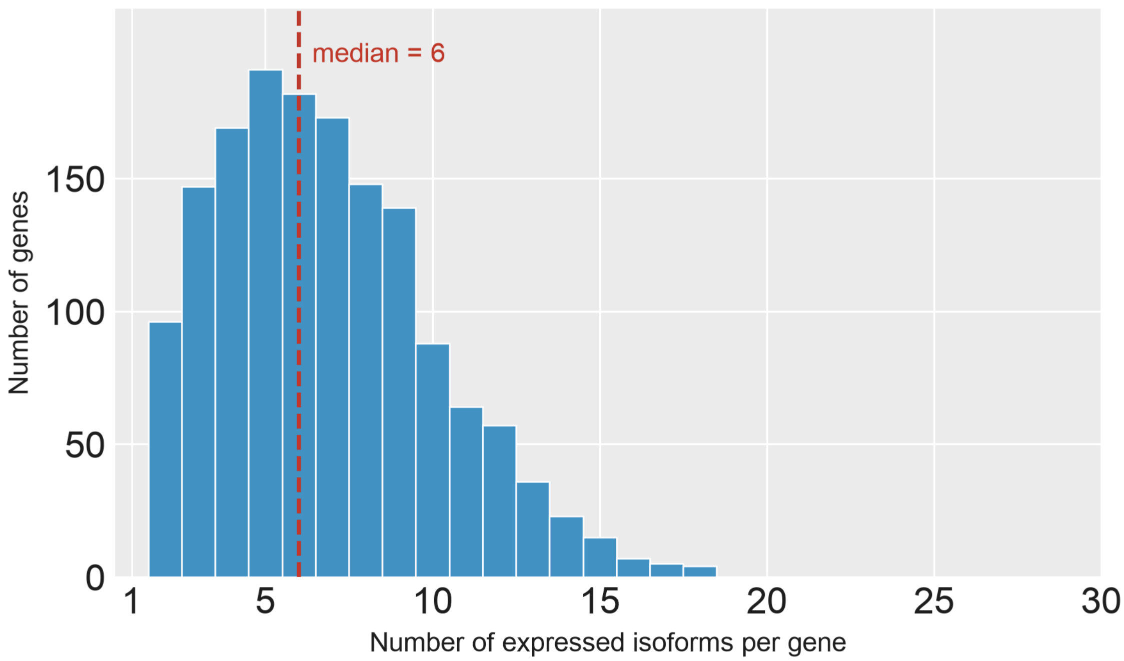
SF24 | Reproducibility Tests of the Productive Response. Four reproducibility diagnostics, sign-flip permutation, cross-donor sign concordance, effect-size variance ratio, and Storey pi0 non-null fraction, shown per cell type (AT2, LAE, FB, MV) against the pass threshold indicated in each row label.



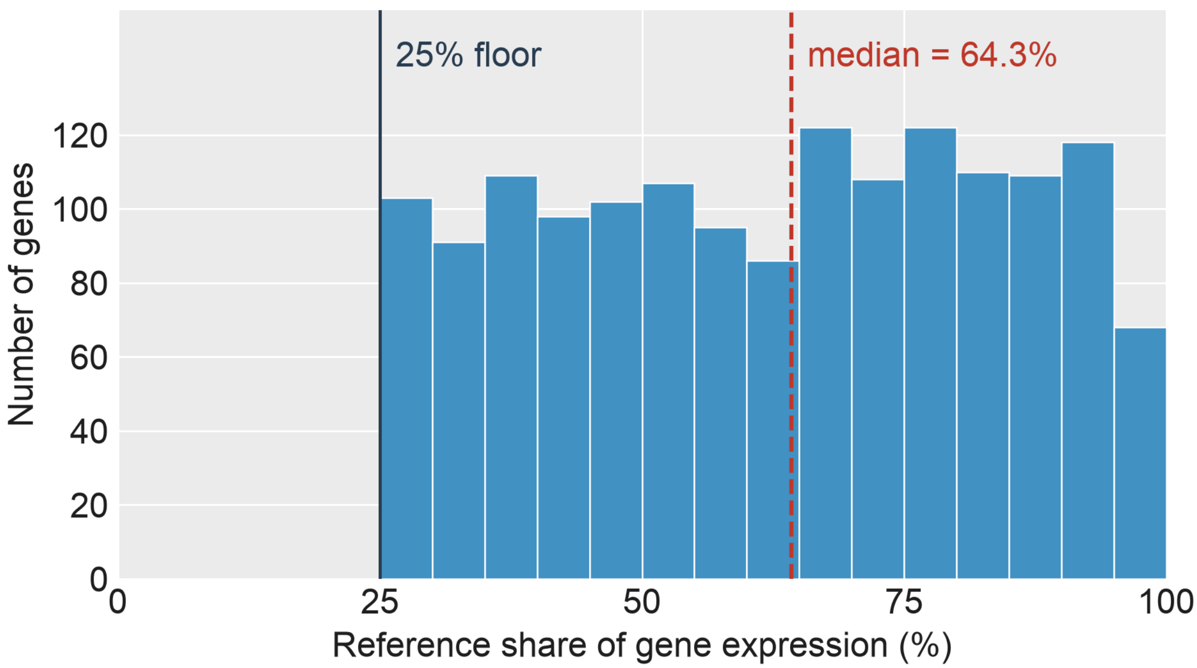
SF25 | Isopair Splice Events. Schematic of the twelve splice-event types identified when comparing a paired isoform (Comparator, bottom track) to the gene's dominant non-NMD isoform (Reference, top track). Gray boxes, exons; horizontal lines with chevrons, introns (chevrons point in the direction of transcription, 5′ → 3′). Categories: transcript-end variants (Alt TSS, Alt TES); alternative splice sites (A5SS, A3SS); exon skipping (SE, Missing Internal); and intron retention in six forms (Partial IR 5′, Partial IR 3′, Full IR, IR diff 5′, IR diff 3′, IR diff 5′ + 3′).



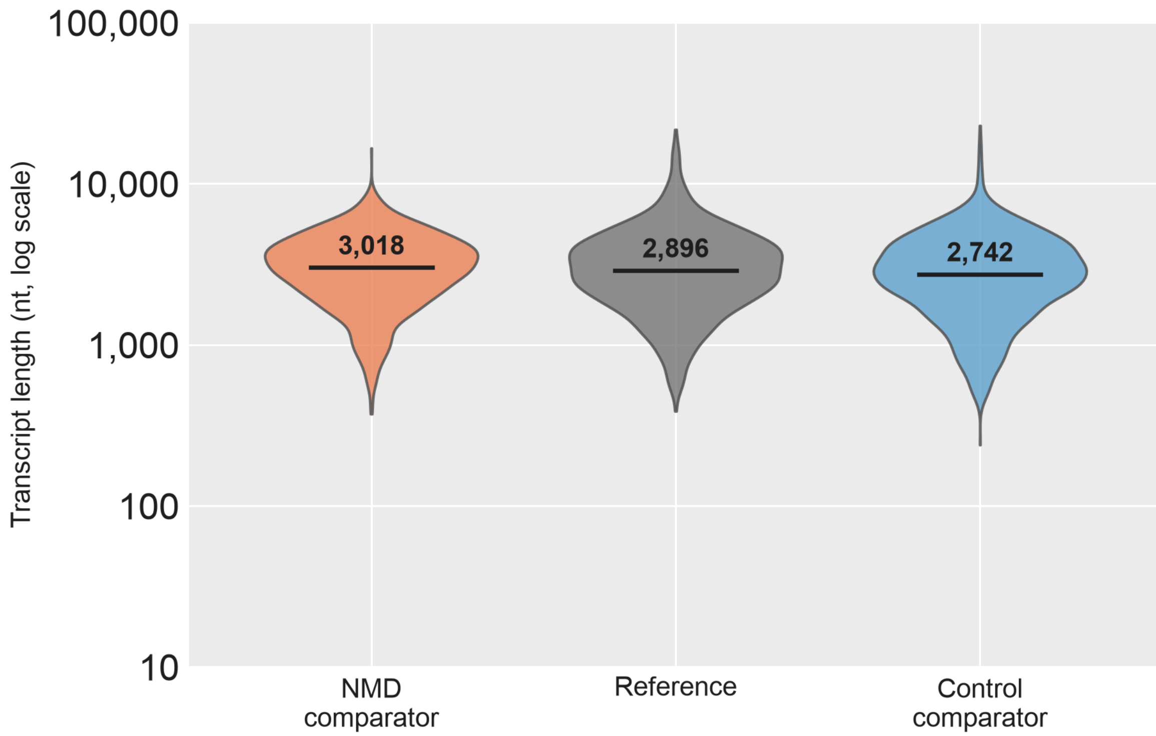
SF26 | Isopair Analysis Data Flow. Data flow for the isoform pairs analyses. Every filter step shows the number of pairs kept and dropped, and a one-line description of why each dropout happened.



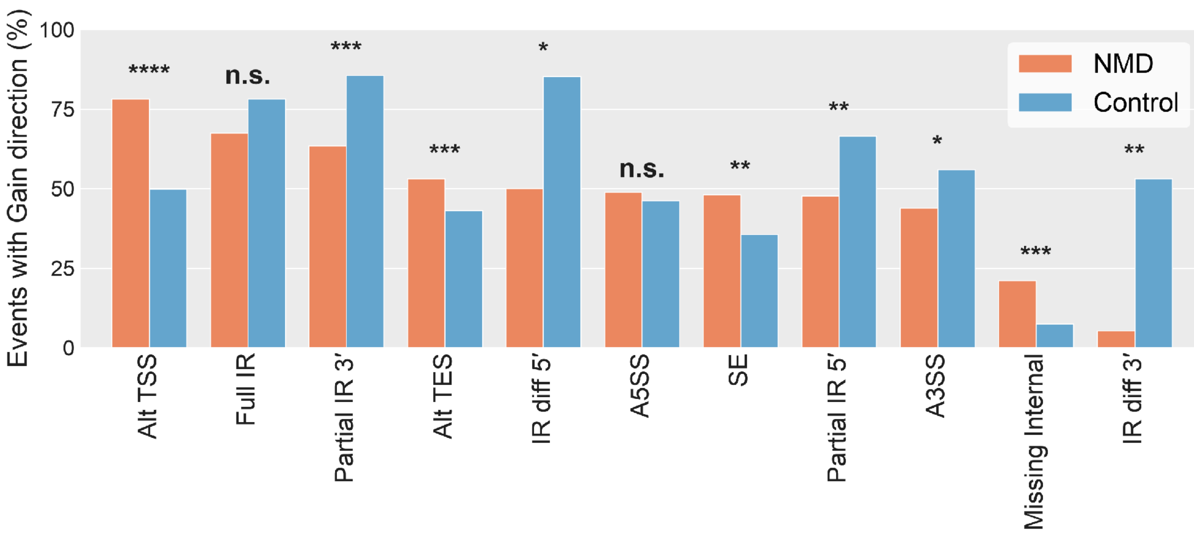
SF27 | Isoform Count in Isoform Pair Sets. Distribution of the number of expressed isoforms per gene across the 1,548 genes for which gene-matched NMD susceptible / reference / non-NMD isoform triplets could be constructed. The median gene contributes 6 isoforms (dashed line). Counts include all long-read isoforms passing the expression filter in at least one of the four cell types (AT2, LAE, FB, MV) under DMSO.



SF28 | Reference Isoform Share of Gene Expression. Distribution across 1,548 genes (each contributing a gene-matched NMD susceptible / reference / non-NMD isoform triplet) of the reference isoform's mean DMSO expression as a fraction of the gene's total isoform expression. Reference isoforms are the highest-expressed non-NMD isoform per gene in DMSO (mean across the four cell types). Dashed line, median; solid line, 25% retention floor.

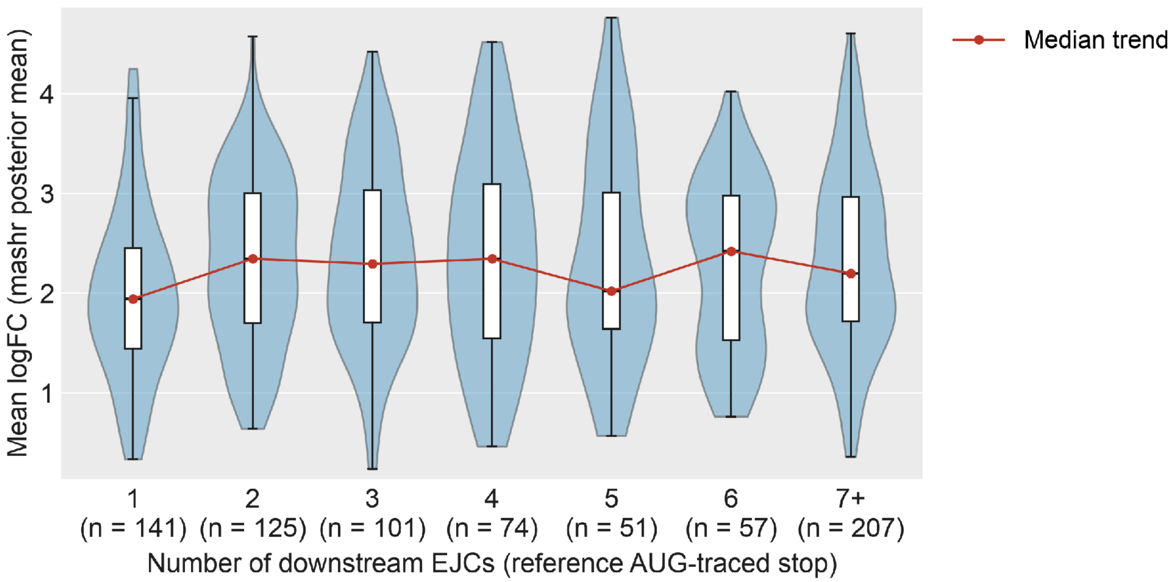


SF29 | Transcript Length Comparison. Violin plot of transcript length (log10 nt) for the 1,548 gene-matched isoform triplets: NMD comparator (orange), Reference (gray), Control comparator (blue). Reference is the highest-expressed non-NMD isoform per gene under DMSO across the four cell types (AT2, LAE, FB, MV); the NMD and Control comparators are the paired NMD susceptible and non-NMD non-reference isoforms from the same gene. Solid black bar, median. Abbreviations. nt, nucleotides.

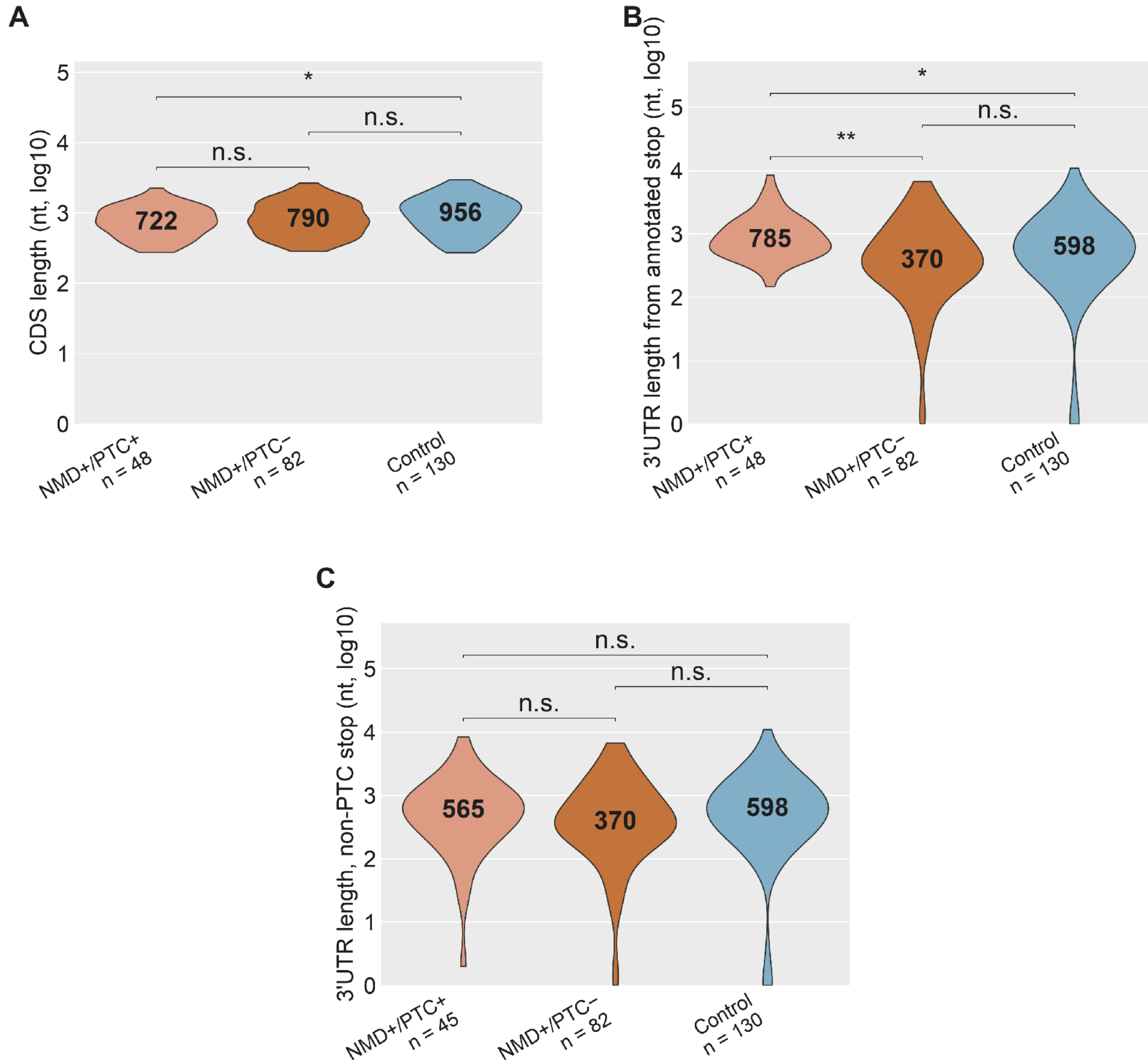


SF30 | Gain Direction by Splice Event Type. Percentage of detected splice events (across 1,548 gene-matched isoform triplets, both NMD and Control comparators) in which the comparator has additional sequence relative to its reference isoform (Gain direction). One bar-pair per event type — NMD comparators (orange) vs Control comparators (blue) — sorted by NMD-side %Gain, descending. Direction is defined from the comparator's perspective (see SF25 for event-type definitions). Significance stars, Fisher's exact test on the 2 × 2 (Gain/Loss × NMD/Control) count table per event type: *$p < 0.05$, **$p < 10^{-3}$, ***$p < 10^{-4}$, ****$p < 10^{-10}$; n.s. otherwise.

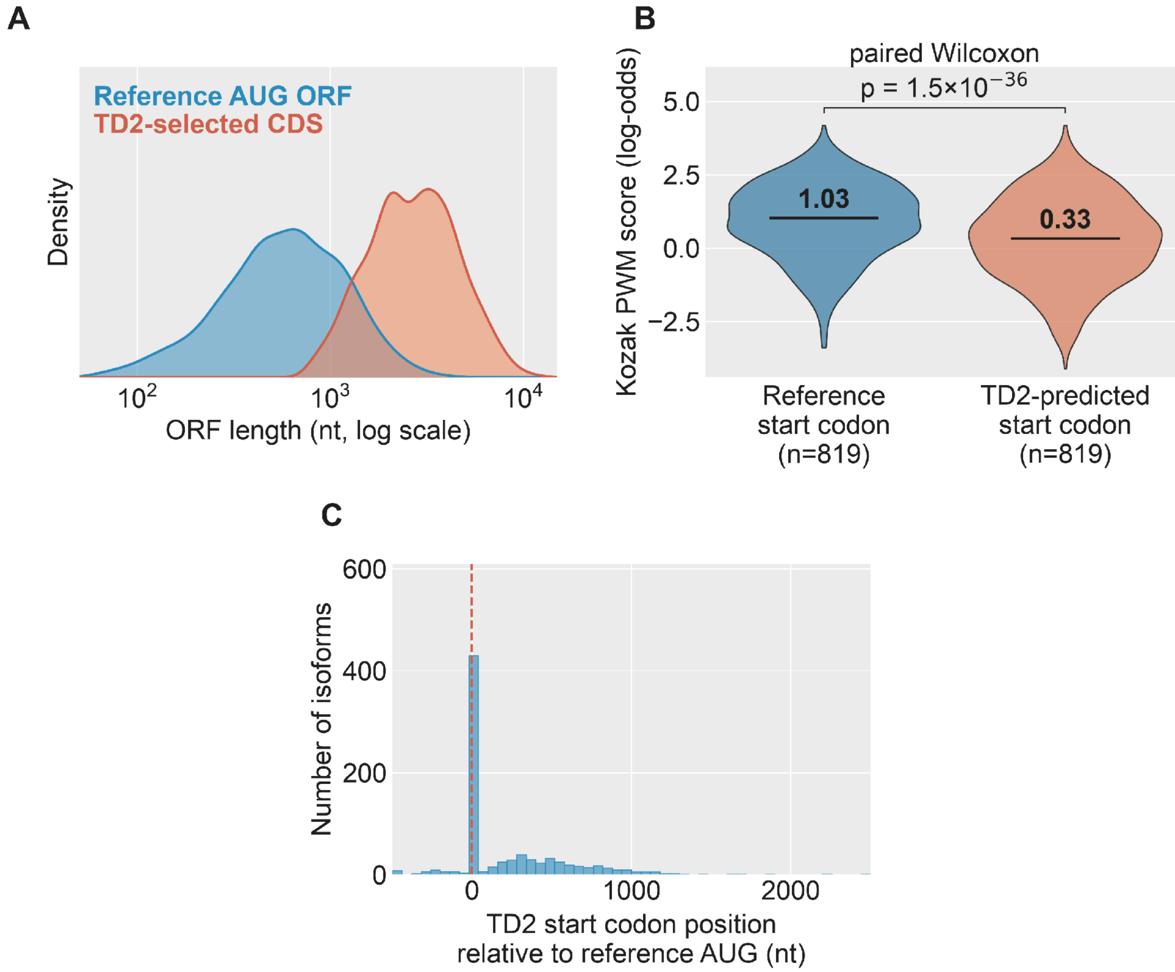
SF31 | NMD-Response Magnitude vs Stop Codon Distance to Last EJC. Per-comparator NMD-response magnitude (mashr posterior-mean $\log_2$ fold-change under SMG1 inhibition, averaged across AT2, LAE, FB and MV) plotted against the distance from the comparator's stop codon to the transcript's last EJC. Positive values, stop codon upstream of the last EJC (candidate NMD substrate); negative values, stop codon downstream of the last EJC. Gray points, individual comparators (1st–99th percentile of x-axis shown); orange line, local-linear smoother; orange band, bootstrap 95% CI. Dashed red line, the operational 50-nt PTC threshold. Population: n = 819 comparators (reference AUG-traceable subset from SF26). Stop anchor: reference AUG-traced (the first in-frame stop reached after projecting the reference gene's AUG onto the comparator); this avoids the attenuation that would arise from TD2 anchoring on non-PTC main ORFs. Median $\log_2$ fold-change is 2.19 above the 50-nt threshold and 1.08 at or below it (Wilcoxon one-sided $p = 3.7 \times 10^{-13}$). Abbreviations. EJC, exon junction complex; nt, nucleotides; ORF, open reading frame.



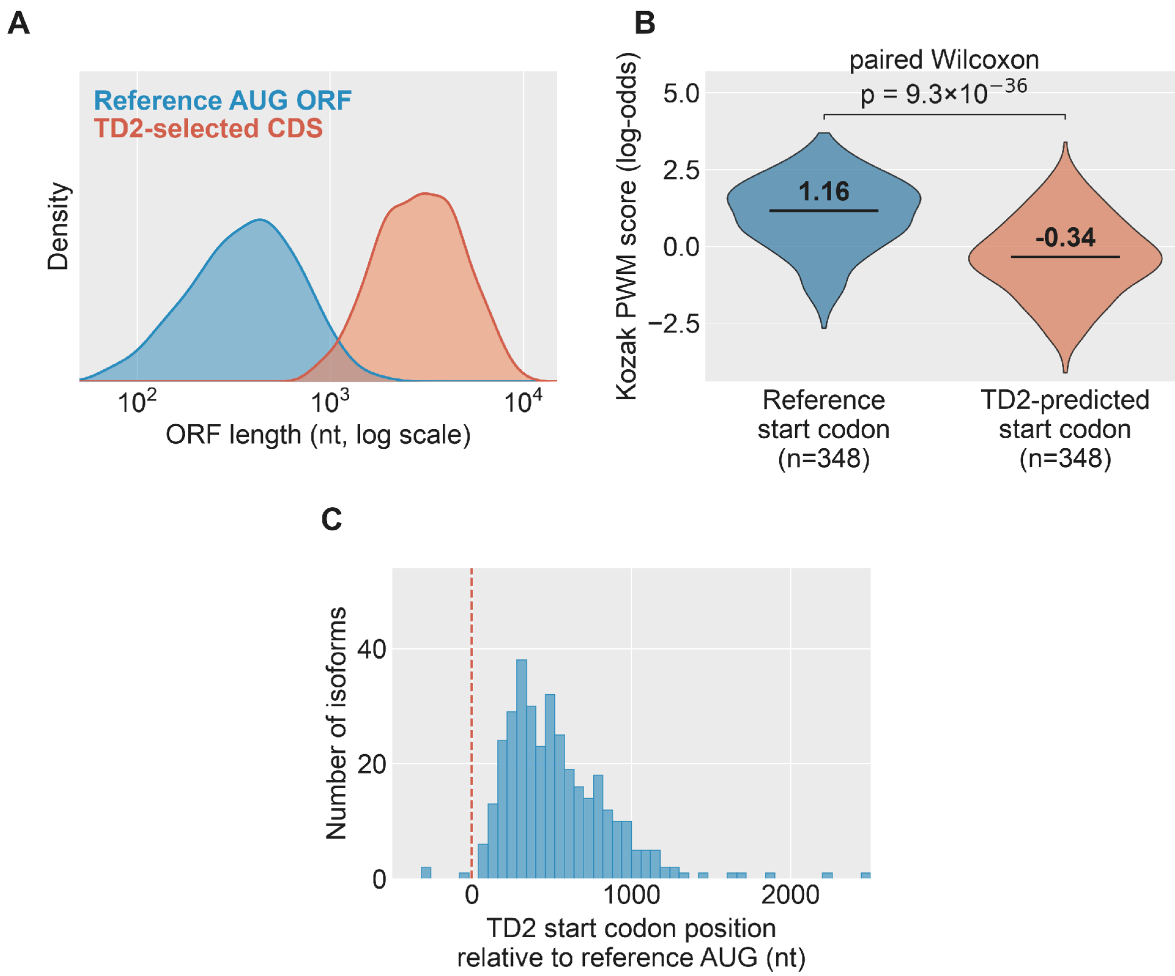
SF32 | NMD Effect Size by Downstream EJC Count in PTC-Positive NMD Comparators. Distribution of NMD-response magnitude (mashr posterior-mean $\log_2$ fold-change under SMG1 inhibition, averaged across AT2, LAE, FB and MV) as a function of the number of EJCs downstream of the comparator's stop codon. Restricted to comparators with at least one downstream EJC (n = 756) from the reference AUG-traceable subset from SF26. Downstream-EJC counts of 7 or more are collapsed into a single "7+" bin. Violin, kernel density; overlaid box, interquartile range and median; red line, per-bin median trend. Per-bin sample counts are shown above each violin. Stop anchor: reference AUG-traced, matching SF31. Abbreviations. EJC, exon junction complex; PTC, premature termination codon; nt, nucleotides.



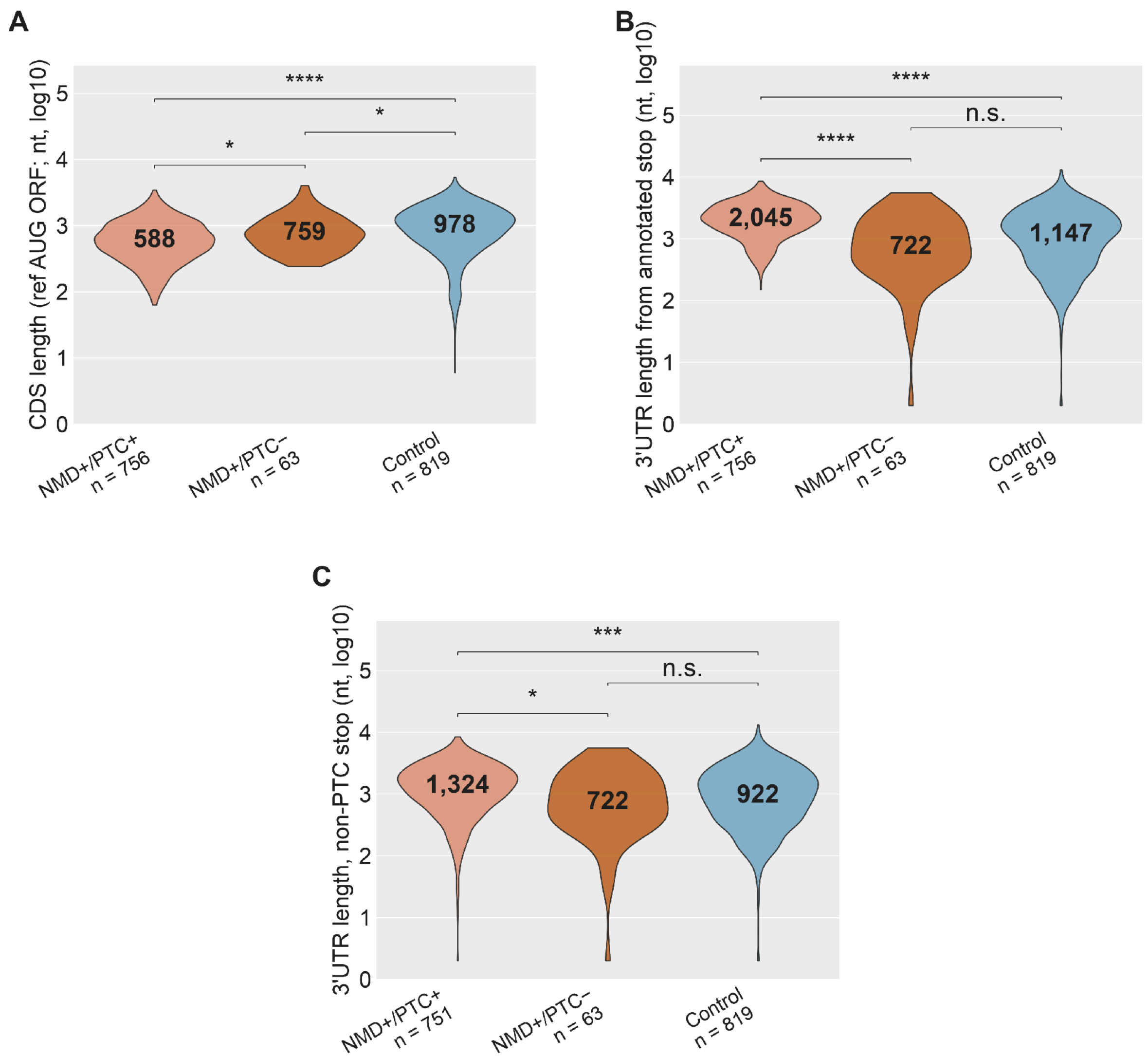
SF33 | CDS and 3′UTR Length Across Isoform Classes in the GENCODE-Restricted Isoform Pairs Subset. Restricted to triplets in which all three transcripts are annotated coding transcripts (n = 48 NMD+/PTC+, 82 NMD+/PTC−, 130 Control). Stop anchor: each isoform's own annotated CDS. (A) CDS length; (B) 3′UTR length from the annotated stop to transcript end; (C) 3′UTR length after walking past any premature stop to the first non-premature stop (Panel C NMD+/PTC+ n = 45; 3 comparators lack a downstream non-premature stop and are dropped). Solid black bar inside each violin, median. Two-sided Wilcoxon rank-sum: *$p < 0.05$, **$p < 10^{-3}$, ***$p < 10^{-4}$, ****$p < 10^{-10}$; n.s. otherwise. Abbreviations. CDS, coding sequence; PTC, premature termination codon.



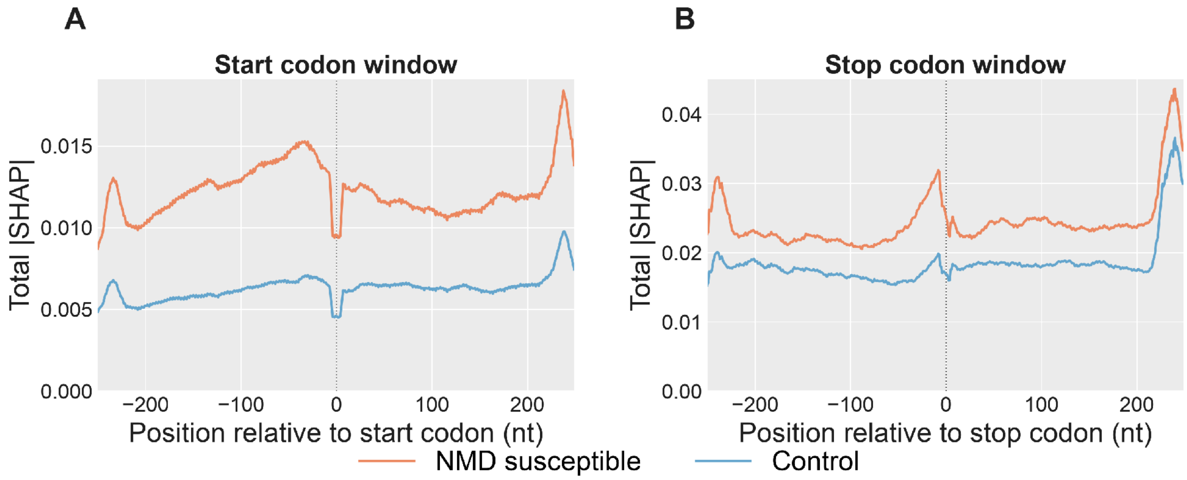
SF34 | TD2 Versus Reference AUG-Traced ORFs, Reference AUG-Traceable Cohort. n = 819 gene-matched NMD comparator pairs. (A) ORF-length distribution under the TD2 call versus under the reference AUG projected onto the same transcript. (B) Paired Kozak PWM score at the reference start codon versus at the TD2-predicted start codon. (C) Position of the TD2 start codon relative to the reference AUG (0 = same site). Solid black bar inside each violin, median. Abbreviations. ORF, open reading frame; PWM, position weight matrix; TD2, TransDecoder2.



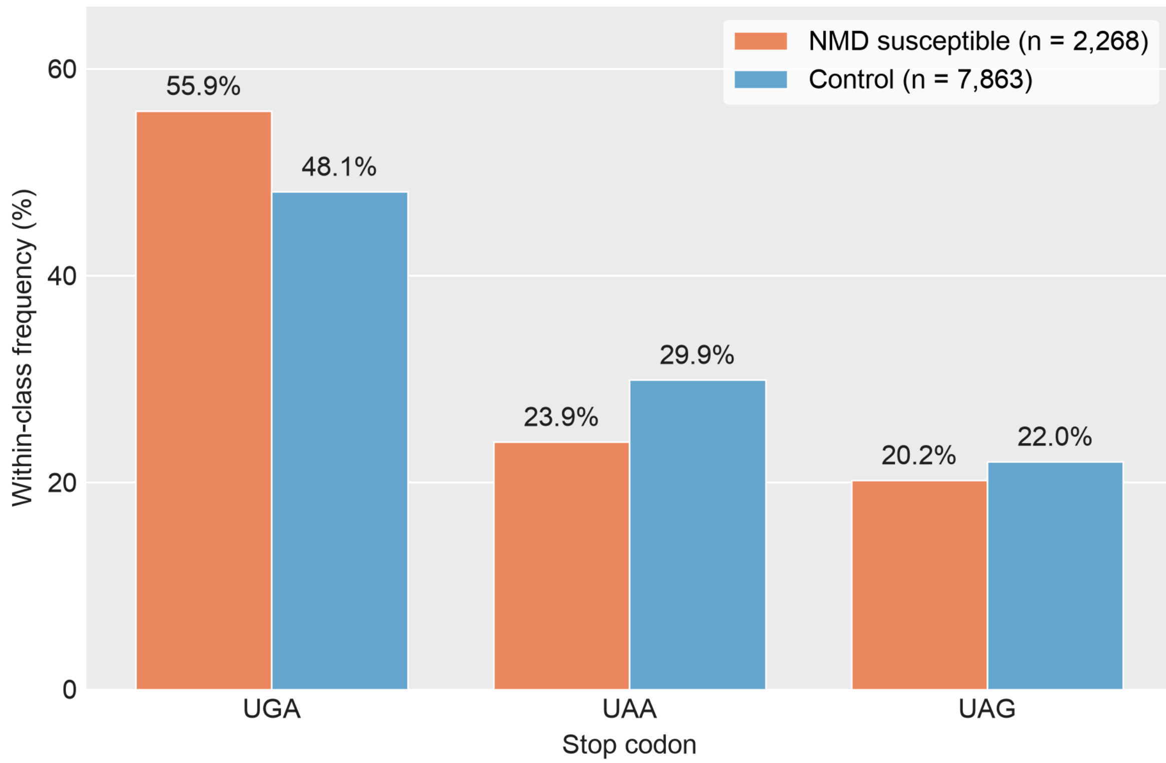
SF35 | TD2 Versus Reference AUG-Traced ORFs, Occult-PTC Subset. Restricted to the n = 348 comparators in which projecting the reference AUG revealed a PTC that the TD2 CDS call did not. (A) ORF-length distribution, TD2 call vs reference AUG projected onto the same transcript. (B) Paired Kozak PWM score at the reference start codon vs the TD2-predicted start codon. (C) Position of the TD2 start codon relative to the reference AUG (0 = same site). Abbreviations. CDS, coding sequence; ORF, open reading frame; PTC, premature termination codon; TD2, TransDecoder2.



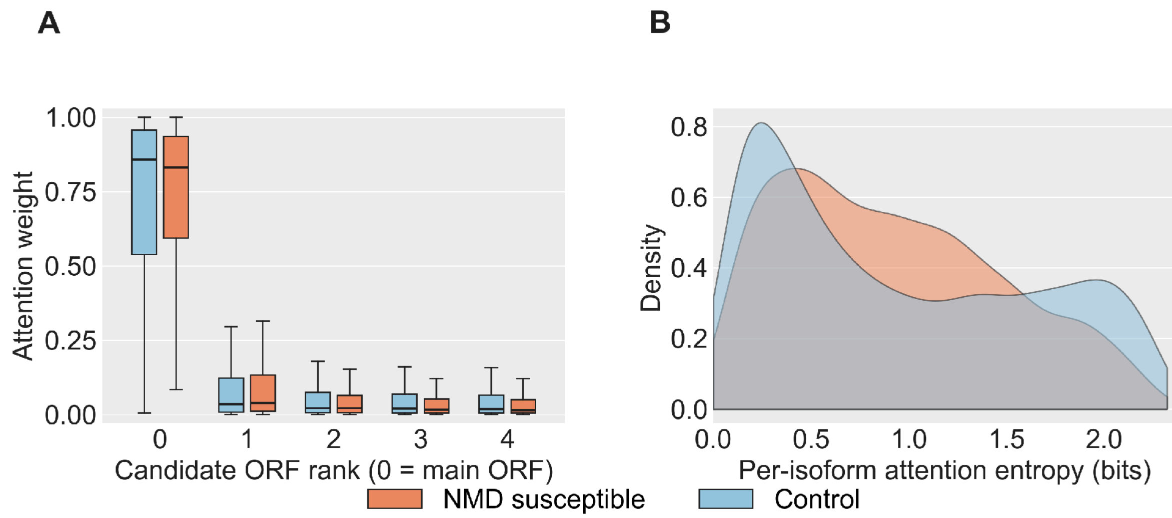
SF36 | CDS and 3′UTR Length Across Isoform Classes, Reference AUG-Traceable Cohort. Restricted to gene-matched pairs in which the reference AUG projects onto the comparator transcript (n = 756 NMD+/PTC+, 63 NMD+/PTC−, 819 Control). Stop anchor: the reference gene's AUG projected onto the comparator, which is unbiased with respect to TD2's PTC-avoidance (see SF34). (A) CDS length; (B) 3′UTR length from the annotated stop; (C) 3′UTR length past the first non-premature stop. Solid black bar inside each violin, median. (Panel C NMD+/PTC+ n = 751; comparators without a downstream non-premature stop are dropped.) Two-sided Wilcoxon rank-sum: *$p < 0.05$, **$p < 10^{-3}$, ***$p < 10^{-4}$, ****$p < 10^{-10}$; n.s. otherwise. Abbreviations. CDS, coding sequence; PTC, premature termination codon; TD2, TransDecoder2.



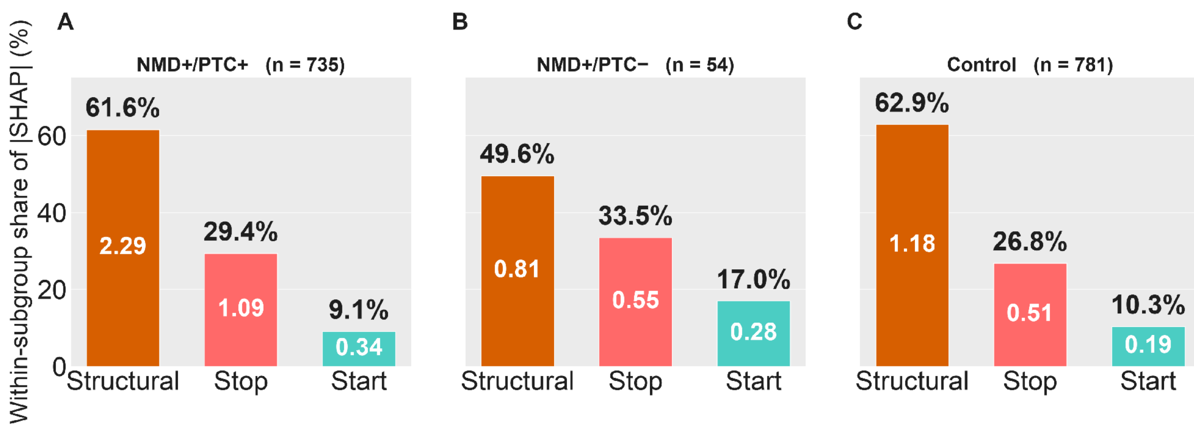
SF37 | Total |SHAP| Across the Start Codon and Stop Codon Windows. Per-position |SHAP| summed across the nine input channels of the deep-learning model, comparing NMD susceptible (orange) and Control (blue) transcripts on the held-out test set (NMD n = 2,268, Control n = 7,863). Each transcript's start codon and stop codon windows are centered on its main ORF's start and stop codons. Main ORFs are chosen with the following priority: the annotated reference CDS (projected from the gene's dominant non-NMD isoform) when available; otherwise the TransDecoder2-called CDS; otherwise the highest-Kozak-scoring ORF (see Methods, Main ORF selection). Values are smoothed with an 11-nt rolling mean over the raw per-position |SHAP| to abstract over trinucleotide-frame periodicity. A) Start codon window covering positions –250 to +249 relative to the start codon. B) Stop codon window covering the same relative range around the stop codon. Pooled across 5 DeepSHAP runs (500 background samples each). Abbreviations. nt, nucleotides; ORF, open reading frame.



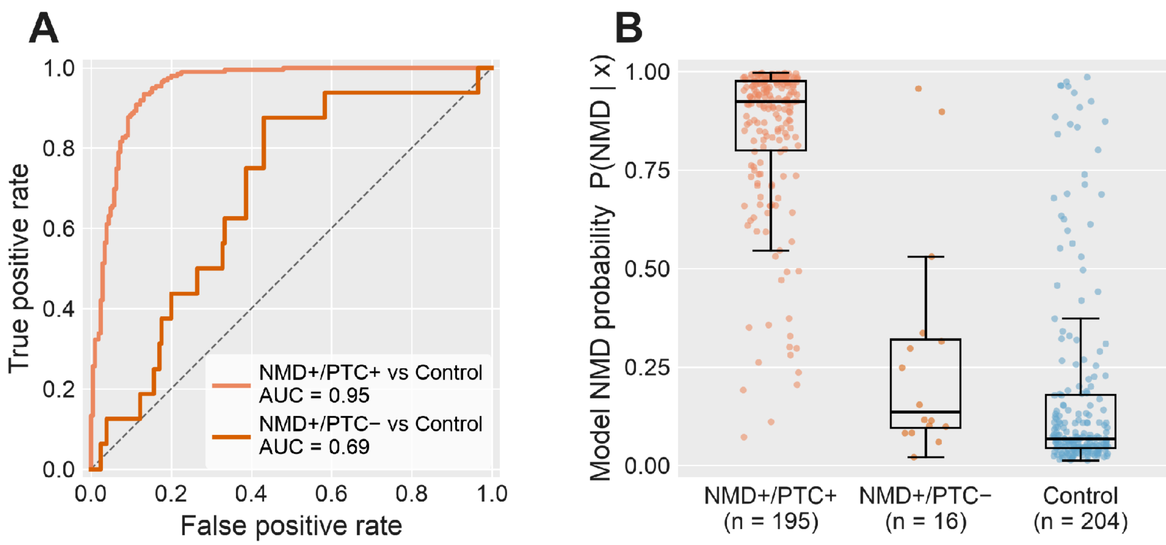
SF38 | Stop Codon Usage in NMD Susceptible vs Non-NMD Isoforms. Within-class frequency of the three canonical stop codons (UGA / UAA / UAG) at the main ORF of each transcript in the deep-learning model's held-out test set (n = 2,268 NMD susceptible, n = 7,863 non-NMD). Main ORFs are chosen with the following priority: the annotated reference CDS (projected from the gene's dominant non-NMD isoform) when available; otherwise the TransDecoder2-called CDS; otherwise the highest-Kozak-scoring ORF (see Methods, Main ORF selection). Percentages: UGA 55.9% vs 48.1%; UAA 23.9% vs 29.9%; UAG 20.2% vs 22.0%. Abbreviations. ORF, open reading frame.



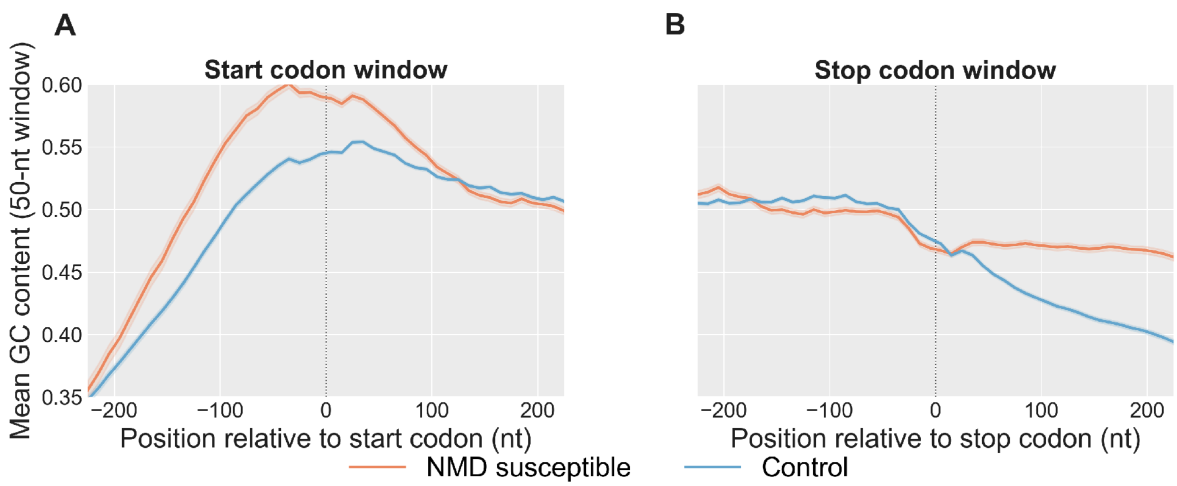
SF39 | Attention Distribution Across the Model's Five Candidate ORFs, NMD Susceptible vs Non-NMD. Held-out test set (NMD n = 2,268; Control n = 7,863). ORF candidates are ranked by the priority described in Methods; rank-0 is the main ORF. (A) Attention weight per candidate ORF rank, by class. (B) Shannon entropy of the per-isoform attention vector across the five candidates; higher entropy indicates more broadly distributed attention. Abbreviations. ORF, open reading frame.



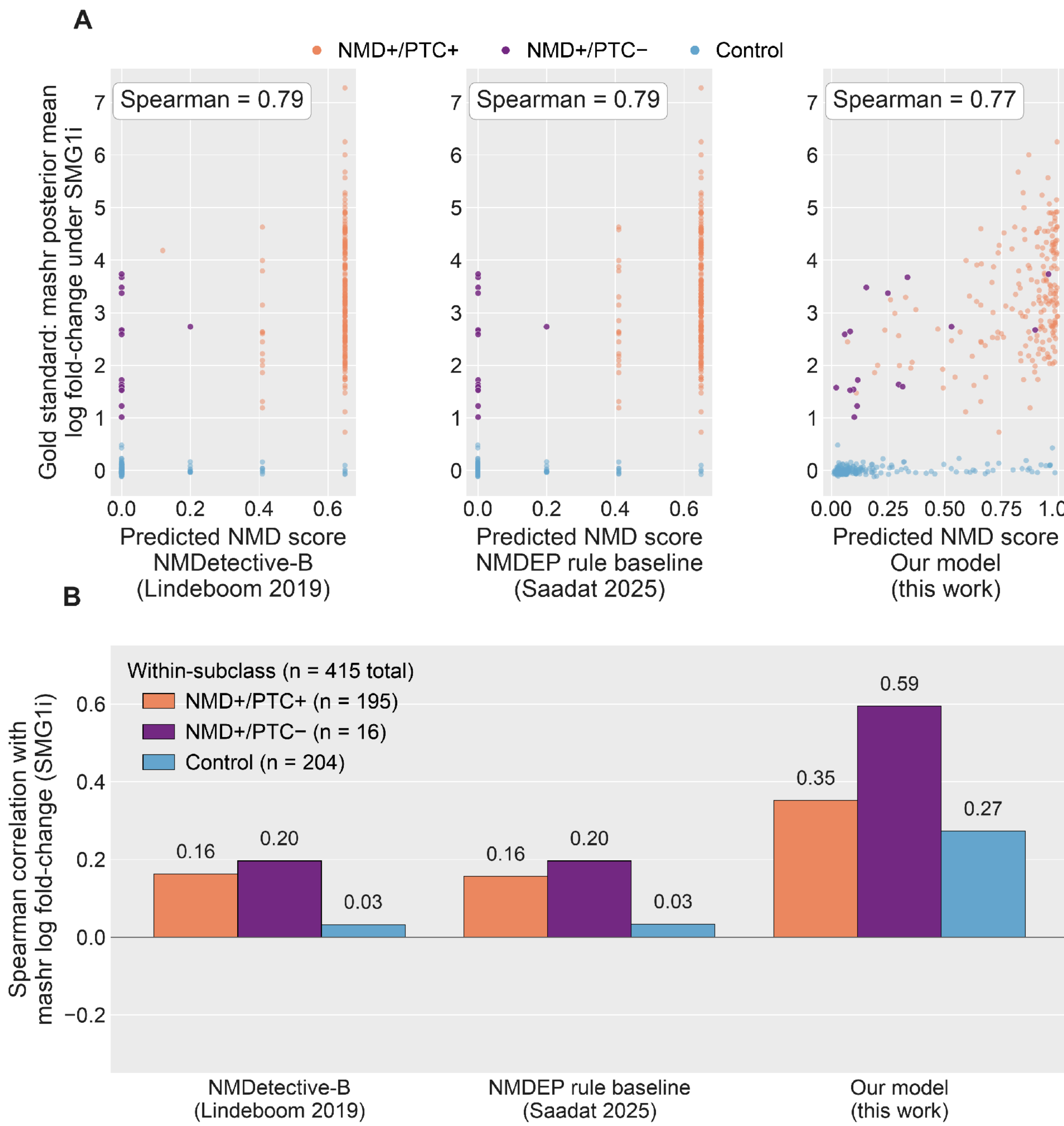
SF40 | Branch-Level Shapley Attributions of the Deep-Learning Model, Stratified by Isoform PTC Subclass. Attributions of the model's NMD prediction across the three input branches (start codon window, stop codon window, structural features). Attributions are shown for the 1,570 of 1,638 reference AUG-traceable isoforms (95.8%) that have a computed model branch attribution, stratified by isoform subclass: NMD+/PTC+ (n = 735), NMD+/PTC− (n = 54), Control (n = 781). The remaining 68 isoforms fall outside the model's scored cohort and are not shown. (A) NMD+/PTC+. (B) NMD+/PTC−. (C) Control. Within each panel, bars are ordered structural / stop / start; the value in each bar is the mean absolute Shapley attribution at that branch; the percentage above each bar is that branch's share of the subclass total. Abbreviations. PTC, premature termination codon.



SF41 | Deep-Learning Model Discrimination by PTC Subclass on the Held-Out Test Set. Held-out test-set performance of the trained NMD predictor, stratified by the three subclasses: NMD+/PTC+ (n = 195), NMD+/PTC− (n = 16), and Control (n = 204). Subgroup sample sizes reflect held-out-test-set membership. (A) ROC curves for the two NMD-vs-Control contrasts, using the model's predicted NMD probability as the score; AUC annotated per contrast; diagonal, chance. (B) Distribution of per-isoform predicted NMD probability by subclass. Abbreviations. AUC, area under the ROC curve; PTC, premature termination codon; ROC, receiver operating characteristic.



SF42 | Rolling GC Content Across the Start Codon and Stop Codon Windows. Mean rolling GC content per position, comparing NMD susceptible (orange) and Control (blue) transcripts on the held-out test set. Windows are centered on each transcript's main ORF start or stop codon; main ORFs are chosen as described in Methods. NMD susceptible transcripts are restricted to those whose main ORF is the annotated reference CDS (n = 1,743 of 2,268 NMD test transcripts, 76.9%), so that position 0 is the reference stop codon; Control transcripts are unrestricted (n = 7,863). GC content is computed in a 50-nt sliding window with a 10-nt step from the model's one-hot nucleotide input channels; shading, ± 1 SEM. (A) Start codon window. (B) Stop codon window. Abbreviations. CDS, coding sequence; GC, guanine + cytosine; nt, nucleotides; ORF, open reading frame; SEM, standard error of the mean.



SF43 | Head-to-Head Correlation of Three NMD Predictors With the mashr Gold Standard, Pooled and per PTC Subclass. Predictors scored against the mashr posterior-mean $\log_2$ fold-change under SMG1 inhibition on the held-out test set of long-read observed transcripts (chromosomes 1, 3, 5, 7; n = 415: NMD+/PTC+ 195, NMD+/PTC− 16, Control 204). Predictors: NMDetective-B (Lindeboom et al. 2019); NMDEP rule-tree baseline (Saadat & Fellay 2025); the deep-learning model. (A) Predicted NMD score versus gold-standard $\log_2$ fold-change, one subplot per model; points colored by subclass; pooled Spearman correlation annotated. (B)

Within-subclass Spearman correlation per model. Abbreviations. mashr, multivariate adaptive shrinkage; NMDEP, NMD efficacy predictor; PTC, premature termination codon.

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